Networking & Data Communications

Complete MCQ Question Bank · Answers & Explanations

OAVS TGT Computer Science Teacher Examination

Exam	OAVS TGT Computer Science Teacher Examination
Subject	Networking and Data Communications
Total Questions	385
Topics	LAN · Communication Concepts · OSI Model · Topologies · Protocols · Ethernet · ARCNET · TCP/IP
Question Types	Definition · Concept · Fact · Application · Analytical · Scenario · Tricky · Assertion-Reason · Match · Statement · Comparison

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Chapter 1: Introduction to LAN

Q1 Definition **LAN**

What does LAN stand for?

A. Local Area Network

B. Large Area Network

C. Long Access Network

D. Local Access Node

Ans: A LAN stands for Local Area Network. It connects devices within a limited geographic area like an office, school, or building.

Q2 Definition **LAN**

A LAN is defined as a network that:

A. Spans multiple cities

B. Connects devices within a limited geographic area

C. Uses satellite communication

D. Covers an entire country

Ans: B A LAN connects computers and devices within a limited area such as a home, office, school, or campus building.

Q3 Concept **LAN**

Which of the following best describes a Local Area Network (LAN)?

A. A network connecting computers across countries

B. A privately owned network within a single building or campus

C. A public network using telephone lines

D. A global computer network

Ans: B LAN is a privately owned network that connects computers and devices within a limited geographic area like a building or campus.

Q4 Characteristic **LAN**

What is the typical geographic coverage of a LAN?

A. Country-wide

B. City-wide

C. Building or campus level

D. Continental

Ans: C LANs typically cover a small geographic area — a single building, floor, office, or campus — usually within a few kilometres.

Q5 Characteristic **LAN**

Which characteristic is true of a LAN?

A. High error rate

B. Low data transfer rate

C. High data transfer rate

D. Uses public switched telephone network

Ans: C LANs offer high data transfer rates (typically 10 Mbps to 10 Gbps) because of short distances and dedicated private infrastructure.

Q6 Characteristic **LAN**

LANs are generally:

A. Owned by government

B. Publicly owned

C. Privately owned

D. Owned by ISPs

Ans: C LANs are typically privately owned by an organization or individual and are confined to a limited geographic area.

Q7 Concept **LAN**

Which of the following is NOT a characteristic of a LAN?

A. Limited geographic coverage

B. High data transfer rate

C. Private ownership

D. Requires satellite communication

Ans: D LANs do not use satellite communication. They use cables or Wi-Fi within a limited area. Satellite communication is used for WANs.

Q8 Advantage **LAN**

Which is an advantage of LAN?

A. Very high installation cost

B. Resource sharing among connected devices

C. Cannot share printers

D. Limited to one device

Ans: B A key advantage of LAN is resource sharing — multiple users can share printers, files, internet connections, and storage devices.

Q9 Advantage **LAN**

LAN allows users to:

A. Access internet only

B. Share files, printers, and resources

C. Use satellite phones

D. Connect only two computers

Ans: B LAN enables resource sharing (files, printers, scanners, internet) among all connected devices, reducing costs and improving efficiency.

Q10 Advantage **LAN**

Which of the following is a benefit of using LAN in an office?

A. Increased hardware cost

B. Easy file sharing and communication

C. Slower processing speed

D. No security

Ans: B LAN facilitates easy file sharing, internal communication, and centralized data storage, making office operations more efficient.

Q11 Component **LAN**

Which device is used to connect multiple devices in a LAN?

A. Router

B. Modem

C. Switch

D. Satellite

Ans: C A switch is a key LAN component that connects multiple devices within the network, forwarding data frames to the correct destination port.

Q12 Component **LAN**

The physical transmission medium commonly used in a LAN is:

A. Satellite link

B. Coaxial cable or twisted pair cable or fiber optic cable

C. Radio waves only

D. Microwave tower

Ans: B LANs commonly use twisted pair cables (Cat5e/Cat6), coaxial cables, or fiber optic cables as the physical transmission medium.

Q13 Component **LAN**

A Network Interface Card (NIC) is used to:

A. Connect a computer to the power supply

B. Connect a computer to a network

C. Increase CPU speed

D. Store data

Ans: B A NIC (Network Interface Card) is hardware installed in a computer that allows it to connect to a network and communicate with other devices.

Q14 Example **LAN**

Which of the following is an example of a LAN?

A. The Internet

B. A network connecting two cities

C. A school computer lab network

D. A nationwide banking network

Ans: C A school computer lab network is a classic example of a LAN — devices connected within a limited geographic area (the lab/building).

Q15 Example **LAN**

An office network connecting 20 computers within one floor of a building is an example of:

A. WAN

B. MAN

C. LAN

D. PAN

Ans: C This is a LAN — a network within a single floor/building, which is within the typical geographic scope of a Local Area Network.

Q16 Concept **LAN**

Which network type covers the smallest geographic area?

A. WAN

B. MAN

C. LAN

D. GAN

Ans: C LAN covers the smallest area — a room, floor, building, or campus. WAN is nationwide/global, MAN covers a city, GAN covers the globe.

Q17 Characteristic **LAN**

The data transfer rate of a typical modern LAN is:

A. 1 bps to 100 bps

B. 1 Kbps to 1 Mbps

C. 10 Mbps to 10 Gbps

D. Only 56 Kbps

Ans: C Modern LANs (Ethernet-based) operate at speeds from 10 Mbps (older) to 1 Gbps or 10 Gbps (Gigabit Ethernet), making them very fast.

Q18 Concept **LAN**

In a LAN, which topology is most commonly used in modern implementations?

A. Bus

B. Ring

C. Star

D. Mesh

Ans: C Star topology is the most commonly used in modern LANs. Each device connects to a central switch/hub, making management and troubleshooting easier.

Q19 Application **LAN**

Which of the following applications does a LAN support?

A. Only email

B. Only file transfer

C. File sharing, printing, communication, and internet access

D. Only internet access

Ans: C LANs support multiple applications: file/printer sharing, internal email, internet access sharing, video conferencing, and centralized storage.

Q20 Comparison **LAN**

Compared to WAN, LAN has:

A. Lower data transfer rate

B. Higher error rate

C. Higher data transfer rate and lower error rate

D. Greater geographic coverage

Ans: C LANs have higher data transfer rates (due to short distances) and lower error rates compared to WANs, which span long distances.

Q21 Tricky **LAN**

Which of the following protocols is primarily associated with LAN communication?

A. HTTP

B. FTP

C. Ethernet (IEEE 802.3)

D. SMTP

Ans: C Ethernet (IEEE 802.3) is the primary protocol standard for LAN communication, defining wiring, signaling, and frame formats.

Q22 Fact **LAN**

The IEEE standard for Ethernet (used in LAN) is:

A. IEEE 802.11

B. IEEE 802.3

C. IEEE 802.15

D. IEEE 802.16

Ans: B IEEE 802.3 is the standard for Ethernet, the most widely used LAN technology. IEEE 802.11 is for Wi-Fi (WLAN).

Q23 Definition **LAN**

A peer-to-peer LAN is one where:

A. All data passes through a central server

B. Each computer can act as both client and server

C. Only one computer has internet access

D. All computers are read-only

Ans: B In a peer-to-peer LAN, every computer can share its resources and act as both a client (requesting) and server (providing) without a dedicated server.

Q24 Concept **LAN**

Which device is used at the boundary of a LAN to connect it to the internet?

A. Switch

B. Hub

C. Router

D. Repeater

Ans: C A router connects a LAN to an external network (like the internet), routing data packets between the two networks using logical addresses.

Q25 Application **LAN**

In which scenario is a LAN most appropriate?

A. Connecting offices in different continents

B. Connecting computers in a single school building

C. Connecting cities across a country

D. Satellite TV broadcasting

Ans: B A LAN is ideal for connecting computers within a single building or campus like a school, where devices are close together.

Q26 Analytical **LAN**

Which type of LAN does NOT require cables?

A. Wired LAN

B. Fiber LAN

C. WLAN (Wireless LAN)

D. Coaxial LAN

Ans: C WLAN (Wireless LAN) uses radio waves (Wi-Fi/IEEE 802.11) instead of cables to connect devices, offering mobility within a limited area.

Q27 Fact **LAN**

Which protocol is used for automatic IP address assignment in a LAN?

A. FTP

B. HTTP

C. DHCP

D. SMTP

Ans: C DHCP (Dynamic Host Configuration Protocol) automatically assigns IP addresses, subnet masks, and gateway addresses to devices joining a LAN.

Q28 Comparison **LAN**

A MAN (Metropolitan Area Network) differs from a LAN primarily in:

A. Ownership model

B. Geographic coverage — MAN covers a city, LAN covers a building

C. Speed

D. Protocol used

Ans: B MAN covers a city or large campus (up to 50 km), while LAN covers a smaller area like a single building or floor.

Q29 Tricky **LAN**

Which transmission medium offers the highest immunity to electromagnetic interference (EMI) in a LAN?

A. Twisted pair cable

B. Coaxial cable

C. Fiber optic cable

D. Radio waves

Ans: C Fiber optic cable transmits light, not electrical signals, making it completely immune to electromagnetic interference (EMI) and RFI.

Q30 Statement **LAN**

Consider: I. LAN covers a small geographic area. II. LAN uses high data transfer rates. III. LAN always requires wireless connections. Which are correct?

A. I only

B. I and II

C. II and III

D. I, II, and III

Ans: B Statements I and II are correct. Statement III is wrong — LAN can use wired (Ethernet) OR wireless connections.

Q236 Fact **LAN**

Which of the following is a server-based LAN?

A. Peer-to-peer network

B. Client-server network

C. Mesh network

D. Ad-hoc network

Ans: B In a client-server LAN, a dedicated server provides resources/services to client computers. This is the standard model in offices and enterprises.

Q237 Concept **LAN**

The maximum cable length for a standard 10Base-T LAN segment is:

A. 500 metres

B. 185 metres

C. 100 metres

D. 50 metres

Ans: C 10Base-T (Cat5 UTP) supports a maximum segment length of 100 metres between a device and the switch/hub. Beyond this, a repeater or switch is needed.

Q238 Concept **LAN**

Which of the following is used to extend the reach of a LAN?

A. Modem

B. Repeater / Switch

C. Satellite dish

D. PBX

Ans: B Repeaters and switches regenerate and forward signals, extending the effective reach of a LAN beyond the single-segment cable length limit.

Q239 | Advantage | **LAN**

Cost reduction in a LAN is achieved because:

A. Each computer needs its own printer

B. Resources like printers and internet connections are shared

C. Cables are very cheap

D. No servers are needed

Ans: B Resource sharing is the primary cost-saving benefit of LAN — one printer, scanner, internet link, or storage device can serve many users simultaneously.

Q240 | Analytical | **LAN**

In a LAN, which layer of the OSI model is primarily responsible for physical addressing?

A. Physical Layer

B. Network Layer

C. Data Link Layer

D. Transport Layer

Ans: C The Data Link Layer uses MAC (physical) addresses to identify devices on a LAN segment and deliver frames between directly connected nodes.

Q241 | Concept | **LAN**

A hub differs from a switch in a LAN because:

A. A hub uses MAC addresses to forward frames

B. A hub broadcasts data to all ports; a switch forwards only to the correct port

C. A hub is more expensive

D. A hub operates at Layer 3

Ans: B A hub (Layer 1) broadcasts all incoming data to every port, creating one collision domain. A switch (Layer 2) uses MAC addresses to forward only to the correct destination port.

Q242	Fact	LAN

Which cable category is most commonly used for Gigabit Ethernet in modern LANs?

A. Cat3

B. Cat5

C. Cat5e / Cat6

D. Cat1

Ans: C Cat5e (enhanced Cat5) and Cat6 cables are the standard for Gigabit Ethernet (1000Base-T) in modern LAN installations, supporting 1 Gbps up to 100 metres.

Q243	Concept	LAN

In a LAN, the term 'bandwidth' refers to:

A. The length of the cable

B. The data transfer capacity of the network

C. The number of connected devices

D. The operating system used

Ans: B Bandwidth in a LAN context refers to the data transfer capacity — the maximum rate at which data can be transmitted, measured in Mbps or Gbps.

Q244	Application	LAN

Which of the following is an internal LAN service used for communication within an organization?

A. Public internet

B. Intranet

C. Extranet

D. Cloud storage

Ans: B An Intranet is a private internal network (based on internet technologies) used within an organization for internal communication and information sharing.

Q245	Tricky	LAN

A LAN that uses no dedicated server and connects 5 computers for file sharing is called:

A. Client-server network

B. Mainframe network

C. Peer-to-peer network

D. Token Ring network

Ans: C A peer-to-peer (P2P) network has no dedicated server — each computer shares resources directly with others. Suitable for small networks (typically under 10 computers).

Chapter 2: Basic Communication Concepts

Q31 Definition **Basic Communication**

Data communication is defined as:

A. Storing data on a hard disk

B. Exchange of data between two devices via a transmission medium

C. Displaying data on a screen

D. Printing documents

Ans: B Data communication is the exchange of data between two or more devices via a transmission medium such as wire, cable, or wireless channel.

Q32 Component **Basic Communication**

Which of the following is NOT a component of a data communication system?

A. Sender

B. Receiver

C. Transmission Medium

D. Power Supply Unit

Ans: D The five basic components of data communication are: Sender, Receiver, Message, Transmission Medium, and Protocol. Power Supply is not one.

Q33 Component **Basic Communication**

In data communication, the 'sender' is:

A. The device that receives the message

B. The device that initiates the message

C. The path through which data travels

D. The rules for communication

Ans: B The sender is the device (computer, phone, etc.) that initiates and sends the data/message in a communication system.

Q34 Component **Basic Communication**

The 'receiver' in a data communication system is:

A. The device that sends data

B. The medium through which data travels

C. The device that accepts the incoming message

D. The protocol used

Ans: C The receiver is the device that receives the transmitted message/data sent by the sender.

Q35 Component **Basic Communication**

The 'message' in data communication refers to:

A. The hardware used

B. The rules of communication

C. The actual data (text, numbers, images, audio, video) being communicated

D. The physical path of data

Ans: C The message is the actual data/information being communicated — it can be text, numbers, pictures, audio, video, or any combination.

Q36 Component **Basic Communication**

The transmission medium in data communication is:

A. The software used to send data

B. The physical path through which data travels from sender to receiver

C. The message being sent

D. The receiver device

Ans: B The transmission medium (channel) is the physical path — wire, cable, fiber optic, or wireless — through which the message travels.

Q37 Definition **Basic Communication**

A protocol in data communication is:

A. A type of cable

B. A set of rules governing data communication

C. A type of modem

D. A network device

Ans: B A protocol is a set of rules and conventions that govern how data is transmitted and received between communicating devices.

Q38 Characteristic **Basic Communication**

Which of the following is a characteristic of effective data communication?

A. Delivery of data to the wrong address

B. Accuracy of transmitted data

C. High error rate

D. Slow speed

Ans: B Accuracy is a key characteristic — data must be delivered exactly as sent without corruption or loss for communication to be effective.

Q39 Characteristic **Basic Communication**

The characteristic 'timeliness' in data communication means:

A. Data is delivered at the right time

B. Data is delivered at any time

C. Data is never delivered

D. Data is delayed

Ans: A Timeliness means data must be delivered within a required time — late delivery (especially in real-time apps like video calls) is useless.

Q40 Characteristic **Basic Communication**

Which characteristic ensures that data is delivered to the correct recipient?

A. Accuracy

B. Timeliness

C. Delivery

D. Jitter control

Ans: C Delivery is the characteristic that ensures data reaches the correct destination — the intended recipient and no one else.

Q41 Concept **Basic Communication**

Simplex communication is:

A. Two-way simultaneous communication

B. One-way communication only

C. Two-way but not simultaneous

D. Multi-directional communication

Ans: B Simplex communication is one-directional — data flows in only one direction. Example: keyboard to computer, traditional TV broadcasting.

Q42 Concept **Basic Communication**

In half-duplex communication:

A. Data flows in one direction only

B. Data flows in both directions simultaneously

C. Data flows in both directions but not simultaneously

D. No data flows

Ans: C Half-duplex allows two-way communication, but only one direction at a time. Example: walkie-talkie.

Q43 Concept **Basic Communication**

Full-duplex communication allows:

A. One-way communication

B. Two-way communication but not simultaneously

C. Two-way simultaneous communication

D. No communication

Ans: C Full-duplex allows simultaneous two-way communication. Example: telephone — both parties can speak and hear at the same time.

Q44 Example **Basic Communication**

Which of the following is an example of simplex communication?

A. Telephone call

B. Walkie-talkie

C. Television broadcasting

D. Video conferencing

Ans: C Television broadcasting is simplex — signal flows only from the broadcaster to the viewer; the viewer cannot send back.

Q45 Example **Basic Communication**

A walkie-talkie is an example of:

A. Simplex

B. Half-duplex

C. Full-duplex

D. Multiplex

Ans: B A walkie-talkie is half-duplex — only one person can transmit at a time; the other must wait and listen.

Q46 Example **Basic Communication**

A telephone conversation is an example of:

A. Simplex

B. Half-duplex

C. Full-duplex

D. No duplex

Ans: C A telephone is full-duplex — both parties can speak and listen simultaneously without switching modes.

Q47 Concept **Basic Communication**

Bandwidth in data communication refers to:

A. The length of a cable

B. The amount of data that can be transmitted in a given time

C. The number of computers

D. The speed of a CPU

Ans: B Bandwidth is the capacity of a transmission medium — the amount of data (measured in bps) that can be transmitted per unit time.

Q48 Concept **Basic Communication**

Noise in data communication is:

A. Useful data signal

B. Any unwanted signal that interferes with the message

C. The speed of transmission

D. The protocol used

Ans: B Noise is any unwanted signal that degrades or distorts the original data signal during transmission, causing errors.

Q49 Concept **Basic Communication**

Attenuation in communication refers to:

A. Increase in signal strength

B. Loss of signal strength over distance

C. Increase in noise

D. Increase in bandwidth

Ans: B Attenuation is the gradual weakening of a signal as it travels over a distance through the transmission medium.

Q50 Concept **Basic Communication**

Which transmission medium uses light pulses to transmit data?

A. Twisted pair cable

B. Coaxial cable

C. Fiber optic cable

D. Radio wave

Ans: C Fiber optic cable transmits data as pulses of light through glass or plastic fibers, offering very high bandwidth and low signal loss.

Q51 Characteristic **Basic Communication**

Which is the fastest transmission medium?

A. Twisted pair cable

B. Coaxial cable

C. Fiber optic cable

D. Wireless radio

Ans: C Fiber optic cable is the fastest transmission medium, supporting extremely high data rates (up to terabits per second) with minimal signal loss.

Q52 Concept **Basic Communication**

Serial transmission means:

A. Data bits are sent simultaneously over multiple channels

B. Data bits are sent one at a time over a single channel

C. Data is sent in parallel

D. No data is sent

Ans: B In serial transmission, bits are sent one after another over a single channel. It is used in most network communications.

Q53 Application **Basic Communication**

Analog signals represent data as:

A. Discrete 0s and 1s

B. Continuous waveforms

C. Digital pulses

D. Binary codes only

Ans: B Analog signals are continuous waveforms that vary in amplitude or frequency. They represent data in a continuous range of values.

Q54 Application **Basic Communication**

Digital signals represent data as:

A. Continuous waveforms

B. Discrete binary values (0 and 1)

C. Analog waves

D. Sound waves

Ans: B Digital signals use discrete binary values (0s and 1s) to represent data. Computers use digital signals for processing and communication.

Q55 Concept **Basic Communication**

A modem is used to:

A. Store data permanently

B. Convert digital signals to analog and vice versa

C. Print documents

D. Connect to a LAN only

Ans: B A modem (MOdulator-DEModulator) converts digital signals from a computer to analog for telephone lines and vice versa.

Q56 Fact **Basic Communication**

The unit of data transmission rate is:

A. Hertz

B. Bits per second (bps)

C. Watts

D. Ohms

Ans: B Data transmission rate is measured in bits per second (bps) or its multiples: Kbps, Mbps, Gbps.

Q57 Tricky **Basic Communication**

Which of the following is a guided (wired) transmission medium?

A. Radio waves

B. Infrared

C. Microwave

D. Fiber optic cable

Ans: D Guided media are physical (wired) media — twisted pair, coaxial, fiber optic. Radio, infrared, and microwave are unguided (wireless) media.

Q58 Tricky **Basic Communication**

Which of the following is an unguided (wireless) transmission medium?

A. Twisted pair cable

B. Coaxial cable

C. Fiber optic

D. Microwave

Ans: D Microwave is an unguided (wireless) transmission medium. Unguided media transmit electromagnetic waves through air without a physical conductor.

Q59 Comparison **Basic Communication**

Compared to wired media, wireless media generally has:

A. Higher security and less interference

B. Lower mobility but more security

C. Greater mobility but more susceptibility to interference

D. Lower cost with equal security

Ans: C Wireless media offers greater mobility (no cables needed) but is more susceptible to interference from other signals and environmental factors.

Q60 Concept **Basic Communication**

Multiplexing in communication refers to:

A. Sending one signal at a time

B. Combining multiple signals to share a single medium

C. Increasing signal noise

D. Reducing bandwidth

Ans: B Multiplexing combines multiple signals for simultaneous transmission over a single medium, improving efficiency. Types: TDM, FDM, WDM.

Q61 Fact **Basic Communication**

TDM stands for:

A. Total Data Multiplexing

B. Time Division Multiplexing

C. Transfer Data Method

D. Timing Data Modulation

Ans: B TDM (Time Division Multiplexing) divides the transmission time into slots, allocating each slot to a different signal/channel for transmission.

Q62 Fact **Basic Communication**

FDM stands for:

A. Frequency Division Multiplexing

B. Frame Data Multiplexing

C. Full Duplex Modulation

D. Fiber Data Medium

Ans: A FDM (Frequency Division Multiplexing) divides the available bandwidth into multiple frequency bands, each carrying a separate signal simultaneously.

Q63 Concept **Basic Communication**

Which type of error occurs when multiple consecutive bits are corrupted by noise?

A. Single-bit error

B. Burst error

C. Checksum error

D. CRC error

Ans: B A burst error is when multiple consecutive bits are corrupted due to a sustained noise event (e.g., electrical interference, physical damage to cable).

Q64 Concept **Basic Communication**

The Shannon-Hartley theorem relates to:

A. Network topology design

B. Maximum data rate of a channel given bandwidth and noise

C. IP address assignment

D. Routing algorithms

Ans: B Shannon-Hartley theorem defines the theoretical maximum data rate (capacity) of a noisy channel given its bandwidth and signal-to-noise ratio.

Q65 Concept **Basic Communication**

Parallel transmission means:

A. One bit is sent at a time

B. Multiple bits are sent simultaneously over multiple channels

C. Data is sent in serial

D. No data is sent

Ans: B In parallel transmission, multiple bits are sent simultaneously over multiple channels/wires. It is faster but more expensive.

Q246 Concept **Basic Communication**

Which of the following is NOT a goal of data communication?

A. Timely delivery

B. Accurate delivery

C. Delivery to the correct destination

D. Delivery at maximum noise levels

Ans: D Goals of data communication are: accurate delivery, timely delivery, and delivery to the correct destination. Maximizing noise is never a goal.

Q247	Concept	Basic Communication

The term 'jitter' in data communication refers to:

A. Constant delay in packet delivery

B. Variation in packet arrival times

C. Loss of packets

D. Increase in bandwidth

Ans: B Jitter is the variation in the delay (latency) of received packets. High jitter causes problems in real-time applications like VoIP and video conferencing.

Q248	Fact	Basic Communication

Which of the following represents the correct formula for data rate according to Nyquist's theorem (noiseless channel)?

A. Max data rate = 2 × Bandwidth × log■(L)

B. Max data rate = Bandwidth / 2

C. Max data rate = Bandwidth × noise

D. Max data rate = log■(Bandwidth)

Ans: A Nyquist's theorem: Maximum data rate = 2 × Bandwidth × log■(L), where L is the number of signal levels. This applies to noiseless channels.

Q249	Concept	Basic Communication

Crosstalk in a cable is caused by:

A. Signal reflection at cable ends

B. Electromagnetic interference between adjacent wires

C. Cable too short

D. High signal strength

Ans: B Crosstalk occurs when electromagnetic signals from one wire interfere with signals in an adjacent wire. Twisted pair cables reduce crosstalk by twisting the wire pairs.

Q250	Concept	Basic Communication

The purpose of twisting wire pairs in UTP (Unshielded Twisted Pair) cable is to:

A. Increase cable length

B. Reduce electromagnetic interference and crosstalk

C. Increase signal strength

D. Reduce cable cost

Ans: B Twisting wire pairs causes the electromagnetic fields from each wire to cancel each other out, reducing electromagnetic interference (EMI) and crosstalk between pairs.

Q251	Application	Basic Communication

Which device converts a digital signal from a computer into an analog signal for transmission over telephone lines?

A. Router

B. Switch

C. Modem

D. Hub

Ans: C A modem (Modulator-Demodulator) converts digital computer signals to analog for transmission over telephone lines (modulation) and converts incoming analog signals back to digital (demodulation).

Q252	Concept	Basic Communication

Latency in data communication is:

A. Data transfer rate

B. The time delay for data to travel from source to destination

C. Signal strength

D. Number of connected devices

Ans: B Latency (delay) is the time it takes for data to travel from the sender to the receiver. High latency degrades real-time applications like online gaming and video calls.

Q253	Fact	Basic Communication

Which of the following is a characteristic of coaxial cable?

A. Very susceptible to EMI

B. Single copper conductor surrounded by insulation, metallic shield, and outer jacket

C. No shielding

D. Only supports 1 Mbps

Ans: B Coaxial cable has a central copper conductor, surrounded by an insulating layer, a metallic (braided) shield, and an outer plastic jacket. The shield reduces EMI.

Q254	Concept	Basic Communication

WDM (Wavelength Division Multiplexing) is used in:

A. Coaxial cable networks

B. Fiber optic networks

C. Wireless networks

D. Twisted pair cable networks

Ans: B WDM (Wavelength Division Multiplexing) transmits multiple data streams simultaneously over a single fiber optic cable using different wavelengths (colors) of light.

Q255	Fact	Basic Communication

The term 'throughput' in data communication refers to:

A. The theoretical maximum data rate

B. The actual data transfer rate achieved in practice

C. The number of errors

D. The cable length

Ans: B Throughput is the actual amount of useful data successfully delivered per unit time, which is typically less than the theoretical bandwidth due to overhead and errors.

Q256	Concept	Basic Communication

Which of the following best describes 'encoding' in data communication?

A. Compressing data

B. Converting data into signals suitable for transmission

C. Routing data

D. Encrypting data

Ans: B Encoding is the process of converting data (bits) into electrical, optical, or radio signals suitable for transmission over a specific medium.

Q257	Tricky	Basic Communication

A fiber optic cable has a core made of:

A. Copper

B. Aluminium

C. Glass or plastic

D. Steel

Ans: C Fiber optic cables have a core made of ultra-pure glass or plastic, through which light pulses travel. The core is surrounded by a cladding layer that reflects light inward.

Q258	Analytical	Basic Communication

Which of the following correctly pairs the communication mode with its example?

A. Full-duplex — walkie-talkie

B. Simplex — telephone

C. Half-duplex — keyboard to computer

D. Full-duplex — telephone call

Ans: D Full-duplex = telephone (simultaneous two-way). Half-duplex = walkie-talkie (one direction at a time). Simplex = keyboard to computer (one direction only).

Q259	Concept	Basic Communication

The OSI layer that handles the conversion of data into signals for transmission is the:

A. Data Link Layer

B. Network Layer

C. Physical Layer

D. Presentation Layer

Ans: C The Physical Layer is responsible for encoding data into electrical, optical, or radio signals and transmitting them over the physical medium.

Q260	Fact	Basic Communication

Shielded Twisted Pair (STP) differs from Unshielded Twisted Pair (UTP) in that STP:

A. Has no twists

B. Has an additional metallic shielding layer around the pairs

C. Is slower

D. Cannot be used in LANs

Ans: B STP has an additional metallic foil or braided shielding around the wire pairs (and sometimes the whole cable), providing better protection against EMI than UTP.

Chapter 3: OSI 7-Layer Model

Q66 Definition **OSI Model**

OSI stands for:

A. Open Software Interface

B. Open Systems Interconnection

C. Operating System Interface

D. Open Signal Interconnect

Ans: B OSI stands for Open Systems Interconnection. It is a conceptual framework developed by ISO to standardize network communication functions.

Q67 Fact **OSI Model**

The OSI model has how many layers?

A. 4

B. 5

C. 6

D. 7

Ans: D The OSI model has exactly 7 layers: Physical, Data Link, Network, Transport, Session, Presentation, and Application.

Q68 Concept **OSI Model**

Which organization developed the OSI model?

A. IEEE

B. IETF

C. ISO

D. ITU

Ans: C The OSI model was developed by ISO (International Organization for Standardization) in 1984 to provide a universal framework for network communication.

Q69 Layer **OSI Model**

The lowest (first) layer of the OSI model is:

A. Data Link Layer

B. Network Layer

C. Physical Layer

D. Transport Layer

Ans: C Layer 1, the Physical Layer, is the lowest layer. It deals with the physical transmission of raw bits over a medium.

Q70 Layer **OSI Model**

The highest (seventh) layer of the OSI model is:

A. Session Layer

B. Presentation Layer

C. Application Layer

D. Transport Layer

Ans: C Layer 7, the Application Layer, is the topmost layer. It provides network services directly to end-user applications.

Q71 Function **OSI Model**

The Physical Layer of the OSI model is responsible for:

A. Logical addressing

B. Routing data packets

C. Transmission of raw bits over a physical medium

D. Session management

Ans: C The Physical Layer (Layer 1) handles the transmission and reception of raw bitstreams over a physical medium — cables, connectors, voltages, and timing.

Q72 Function **OSI Model**

Which of the following is a function of the Physical Layer?

A. Encryption

B. Flow control

C. Bit synchronization

D. Routing

Ans: C Bit synchronization (ensuring sender and receiver are in sync) is a key Physical Layer function. It also defines encoding and line configuration.

Q73 Device **OSI Model**

Which device operates at the Physical Layer of the OSI model?

A. Switch

B. Router

C. Hub / Repeater

D. Gateway

Ans: C Hubs and repeaters operate at the Physical Layer (Layer 1) — they simply amplify or broadcast raw signals without interpreting them.

Q74 Function **OSI Model**

The Data Link Layer is responsible for:

A. Routing packets between networks

B. Node-to-node delivery of data frames

C. End-to-end data delivery

D. Session establishment

Ans: B The Data Link Layer (Layer 2) ensures reliable node-to-node (hop-to-hop) delivery of data frames between directly connected devices.

Q75 Function **OSI Model**

MAC addressing is a function of which OSI layer?

A. Physical Layer

B. Data Link Layer

C. Network Layer

D. Transport Layer

Ans: B MAC (Media Access Control) addressing is part of the Data Link Layer. MAC addresses identify devices on a LAN segment.

Q76 Concept **OSI Model**

The Data Link Layer divides into two sub-layers:

A. LLC and MAC

B. TCP and UDP

C. HTTP and FTP

D. IP and ICMP

Ans: A The Data Link Layer has two sub-layers: LLC (Logical Link Control) — for flow control and error checking, and MAC (Media Access Control) — for addressing.

Q77 Device **OSI Model**

Which device operates at the Data Link Layer?

A. Hub

B. Router

C. Layer-2 Switch

D. Repeater

Ans: C A Layer 2 switch operates at the Data Link Layer — it uses MAC addresses to forward frames to the correct port within a LAN.

Q78 Function **OSI Model**

The Network Layer is primarily responsible for:

A. Physical transmission of bits

B. Logical addressing and routing

C. Session management

D. Data encryption

Ans: B The Network Layer (Layer 3) handles logical addressing (IP addresses) and routing — determining the best path for data packets across networks.

Q79 Function **OSI Model**

IP addressing is a function of which OSI layer?

A. Data Link Layer

B. Physical Layer

C. Network Layer

D. Transport Layer

Ans: C IP (Internet Protocol) addressing is a Network Layer (Layer 3) function. IP addresses enable logical identification and routing across interconnected networks.

Q80 Device **OSI Model**

Which device operates at the Network Layer?

A. Hub

B. Switch

C. Router

D. Gateway

Ans: C Routers operate at the Network Layer (Layer 3) — they use IP addresses to forward packets between different networks along the best path.

Q81 Function **OSI Model**

The Transport Layer provides:

A. Physical bit transmission

B. Logical addressing

C. End-to-end reliable delivery and error recovery

D. Session dialogue control

Ans: C The Transport Layer (Layer 4) ensures end-to-end reliable delivery of data, error detection/correction, flow control, and segmentation/reassembly.

Q82 Fact **OSI Model**

Which protocols operate at the Transport Layer?

A. HTTP and FTP

B. IP and ICMP

C. TCP and UDP

D. Ethernet and PPP

Ans: C TCP (Transmission Control Protocol) and UDP (User Datagram Protocol) are the primary Transport Layer protocols.

Q83 Function **OSI Model**

Port numbers are associated with which OSI layer?

A. Network Layer

B. Data Link Layer

C. Transport Layer

D. Session Layer

Ans: C Port numbers are used at the Transport Layer to identify specific processes/applications on a host (e.g., port 80 for HTTP, port 443 for HTTPS).

Q84 Function **OSI Model**

The Session Layer is responsible for:

A. Physical transmission

B. Routing

C. Establishing, managing, and terminating sessions

D. Data encryption

Ans: C The Session Layer (Layer 5) establishes, manages, synchronizes, and terminates communication sessions between applications on different devices.

Q85 Function **OSI Model**

Which of the following is a function of the Session Layer?

A. IP addressing

B. Frame synchronization

C. Dialog control and synchronization

D. Bit encoding

Ans: C Dialog control (managing turn-taking in communication) and synchronization (checkpoints for recovery) are key Session Layer functions.

Q86 Function **OSI Model**

The Presentation Layer is responsible for:

A. Routing data

B. Establishing sessions

C. Data translation, encryption, and compression

D. Physical bit transmission

Ans: C The Presentation Layer (Layer 6) handles data translation (format conversion), encryption/decryption, and compression/decompression.

Q87 Function **OSI Model**

Encryption in the OSI model is typically handled at:

A. Physical Layer

B. Network Layer

C. Presentation Layer

D. Transport Layer

Ans: C Encryption and decryption are Presentation Layer (Layer 6) functions — it ensures data is in a secure, usable format for the Application Layer.

Q88 Function **OSI Model**

Data compression is a function of which OSI layer?

A. Application Layer

B. Session Layer

C. Presentation Layer

D. Data Link Layer

Ans: C The Presentation Layer handles data compression to reduce the amount of data that needs to be transmitted across the network.

Q89 Function **OSI Model**

The Application Layer of the OSI model:

A. Handles physical connections

B. Manages routing

C. Provides network services directly to end-user applications

D. Controls sessions

Ans: C The Application Layer (Layer 7) provides network services (email, file transfer, web browsing) directly to end-user applications.

Q90 Example **OSI Model**

Which of the following protocols operates at the Application Layer?

A. TCP

B. IP

C. HTTP

D. Ethernet

Ans: C HTTP (HyperText Transfer Protocol) operates at the Application Layer (Layer 7). It is used for web browsing and data exchange on the internet.

Q91 Mnemonic **OSI Model**

The correct order of OSI layers from bottom to top is:

A. Application, Session, Transport, Network, Data Link, Physical, Presentation

B. Physical, Data Link, Network, Transport, Session, Presentation, Application

C. Network, Physical, Data Link, Transport, Session, Presentation, Application

D. Physical, Network, Data Link, Transport, Session, Presentation, Application

Ans: B From bottom (Layer 1) to top (Layer 7): Physical, Data Link, Network, Transport, Session, Presentation, Application. Mnemonic: 'Please Do Not Throw Sausage Pizza Away'.

Q92 Comparison **OSI Model**

Which OSI layer is responsible for segmentation of data?

A. Network Layer

B. Data Link Layer

C. Transport Layer

D. Presentation Layer

Ans: C The Transport Layer segments large messages into smaller units (segments) and reassembles them at the destination.

Q93 Tricky **OSI Model**

The PDU (Protocol Data Unit) at the Network Layer is called:

A. Frame

B. Segment

C. Packet

D. Bit

Ans: C Network Layer PDU = Packet. Transport Layer PDU = Segment. Data Link Layer PDU = Frame. Physical Layer PDU = Bit.

Q94 Tricky **OSI Model**

The PDU (Protocol Data Unit) at the Data Link Layer is called:

A. Packet

B. Frame

C. Segment

D. Bit

Ans: B The Data Link Layer PDU is called a Frame. It encapsulates the Network Layer packet with MAC addressing, framing, and error checking.

Q95 Tricky **OSI Model**

The PDU at the Transport Layer is called:

A. Bit

B. Frame

C. Packet

D. Segment

Ans: D The Transport Layer PDU is called a Segment (in TCP) or Datagram (in UDP). It carries the payload from the Application/Session layers.

Q96 Application **OSI Model**

Which OSI layer translates ASCII to EBCDIC?

A. Application Layer

B. Presentation Layer

C. Session Layer

D. Transport Layer

Ans: B The Presentation Layer handles character code translation (e.g., ASCII to EBCDIC), ensuring different systems can understand each other.

Q97 Analytical **OSI Model**

At which OSI layer does a Layer-2 switch forward data based on MAC addresses?

A. Physical Layer

B. Network Layer

C. Data Link Layer

D. Transport Layer

Ans: C A Layer 2 switch operates at the Data Link Layer and uses MAC addresses to forward frames to the appropriate port on the local network.

Q98 Analytical **OSI Model**

At which OSI layer does a router make forwarding decisions?

A. Physical Layer

B. Data Link Layer

C. Network Layer

D. Session Layer

Ans: C Routers operate at the Network Layer (Layer 3), using IP addresses and routing tables to determine the best path for forwarding packets.

Q99 Concept **OSI Model**

The process of adding headers as data moves down the OSI layers is called:

A. Decapsulation

B. Encapsulation

C. Routing

D. Segmentation

Ans: B Encapsulation is the process of adding layer-specific headers as data moves from the Application Layer down to the Physical Layer. Decapsulation is the reverse.

Q100 Fact **OSI Model**

How many layers are above the Network Layer in the OSI model?

A. 2

B. 3

C. 4

D. 5

Ans: C Above the Network Layer (Layer 3) are: Transport (4), Session (5), Presentation (6), and Application (7) — that is 4 layers.

Q261 Concept **OSI Model**

Which OSI layer is responsible for end-to-end error recovery?

A. Data Link Layer

B. Network Layer

C. Transport Layer

D. Session Layer

Ans: C The Transport Layer provides end-to-end error recovery — detecting and requesting retransmission of lost or corrupted segments between the source and destination hosts.

Q262 Fact **OSI Model**

The IEEE 802.3 standard relates to which OSI layer(s)?

A. Physical Layer only

B. Data Link Layer only

C. Both Physical and Data Link Layers

D. Network Layer

Ans: C IEEE 802.3 (Ethernet) specifies both Physical Layer (cable types, signaling) and Data Link Layer (MAC addressing, frame format) standards.

Q263 Concept **OSI Model**

Which OSI layer adds trailer information (FCS) to the data?

A. Physical Layer

B. Transport Layer

C. Data Link Layer

D. Network Layer

Ans: C The Data Link Layer adds both a header (with MAC addresses) and a trailer (FCS — Frame Check Sequence) to detect transmission errors in the frame.

Q264 Analytical **OSI Model**

When a router receives a packet, it reads the header up to which OSI layer?

A. Physical Layer

B. Data Link Layer

C. Network Layer

D. Transport Layer

Ans: C A router operates at the Network Layer (Layer 3). It reads up to the Network Layer header (IP address) to make routing decisions, then re-encapsulates the packet.

Q265 Concept **OSI Model**

Which OSI layer is responsible for reliable end-to-end data transfer even if intermediate networks are unreliable?

A. Network Layer

B. Data Link Layer

C. Session Layer

D. Transport Layer

Ans: D The Transport Layer provides reliable end-to-end communication even across unreliable networks, using acknowledgements, sequencing, and retransmission.

Q266 Comparison **OSI Model**

The main difference between the Physical Layer and the Data Link Layer is:

A. Physical Layer uses IP addresses; Data Link uses MAC addresses

B. Physical Layer transmits raw bits; Data Link Layer organizes bits into frames with addressing

C. Physical Layer is faster

D. Data Link Layer uses radio signals

Ans: B Physical Layer: transmits raw bits over a medium. Data Link Layer: organizes bits into frames, adds MAC addresses, and provides error detection for the local link.

Q267 Concept **OSI Model**

Which OSI layer is responsible for establishing, maintaining, and terminating logical connections between applications?

A. Transport Layer

B. Data Link Layer

C. Session Layer

D. Presentation Layer

Ans: C The Session Layer (Layer 5) manages the dialogue between applications — establishing, maintaining, synchronizing, and gracefully terminating communication sessions.

Q268 Tricky **OSI Model**

NFS (Network File System) and SQL operate at which OSI layer?

A. Transport Layer

B. Network Layer

C. Session Layer

D. Application Layer

Ans: D NFS and SQL are application-level protocols/interfaces operating at the Application Layer (Layer 7), providing network-based file and database services to applications.

Q269 Tricky **OSI Model**

Which device works at all seven layers of the OSI model?

A. Hub

B. Switch

C. Router

D. Gateway / Firewall

Ans: D A gateway (and modern firewalls) can operate at all 7 layers — it translates between different protocol stacks and can inspect data at every layer.

Q270 Concept **OSI Model**

The concept of 'virtual circuits' is associated with which OSI layer?

A. Physical Layer

B. Data Link Layer

C. Network Layer

D. Transport Layer

Ans: D Virtual circuits (connection-oriented data transfer paths) are a Transport Layer concept. TCP uses virtual circuits to provide reliable, ordered byte-stream delivery.

Q271 Fact **OSI Model**

Which OSI layer is responsible for handling multiple connections from the same application simultaneously?

A. Network Layer

B. Transport Layer

C. Session Layer

D. Application Layer

Ans: B The Transport Layer uses port numbers to multiplex/demultiplex multiple simultaneous connections from different applications on the same host.

Q272 Application **OSI Model**

SMTP (Simple Mail Transfer Protocol) operates at which OSI layer?

A. Transport Layer

B. Network Layer

C. Data Link Layer

D. Application Layer

Ans: D SMTP is an Application Layer (Layer 7) protocol used for sending email between mail servers and from email clients to mail servers.

Q273 Concept **OSI Model**

The term 'peer-layer communication' in the OSI model means:

A. All layers communicate with each other simultaneously

B. Each layer communicates logically with the same layer on the remote system

C. Layers skip each other

D. Only adjacent layers communicate

Ans: B In OSI, each layer communicates logically (via headers) with its peer layer on the remote system, even though data physically passes through all layers at the sender and receiver.

Q274 Comparison **OSI Model**

Which two OSI layers are combined in the TCP/IP Network Access layer?

A. Physical and Network

B. Network and Transport

C. Physical and Data Link

D. Data Link and Network

Ans: C The TCP/IP Network Access (Link) layer combines the functions of both the OSI Physical Layer (Layer 1) and Data Link Layer (Layer 2).

Q275 Tricky **OSI Model**

Which layer of the OSI model is responsible for converting data formats between different systems (e.g., JPEG, ASCII, MPEG)?

A. Application Layer

B. Session Layer

C. Presentation Layer

D. Transport Layer

Ans: C The Presentation Layer (Layer 6) handles data format conversion (e.g., ASCII to EBCDIC, JPEG image format, MPEG video encoding), ensuring data is readable by the Application Layer.

Q276 Analytical **OSI Model**

A repeater operates at which OSI layer and what does it do?

A. Layer 2 — forwards frames

B. Layer 3 — routes packets

C. Layer 1 — regenerates and amplifies signals

D. Layer 4 — manages sessions

Ans: C A repeater operates at the Physical Layer (Layer 1). It regenerates (amplifies and reshapes) electrical/optical signals to extend the cable distance without interpreting the data.

Q277	Concept	OSI Model

Flow control at the Data Link Layer prevents:

A. Network congestion across routers

B. A fast sender from overwhelming a slow receiver on the same link

C. IP address conflicts

D. Session timeouts

Ans: B Data Link Layer flow control prevents a fast transmitting node from overwhelming a slower receiving node on the same local link, using mechanisms like sliding window or stop-and-wait.

Q278	Fact	OSI Model

Which OSI layer handles syntax and semantics of the information exchanged?

A. Application Layer

B. Session Layer

C. Presentation Layer

D. Transport Layer

Ans: C The Presentation Layer deals with the syntax (format) and semantics (meaning) of data — it ensures data from the Application Layer is presented in a standard, understandable format.

Chapter 4: Network Topologies

Q101	Definition	Network Topology

Network topology refers to:

A. The speed of a network

B. The physical or logical arrangement of devices in a network

C. The number of users on a network

D. The protocol used in a network

Ans: B Network topology is the arrangement or layout of the various elements (nodes, links) of a computer network — both physically and logically.

Q102	Definition	Bus Topology

In a Bus topology:

A. All devices connect to a central hub

B. All devices share a single common communication line (backbone)

C. Every device connects to every other device

D. Devices are arranged in a ring

Ans: B In Bus topology, all devices are connected to a single backbone cable. Data travels along the bus and is received by all nodes, but only the addressed node accepts it.

Q103	Disadvantage	Bus Topology

Which of the following is a disadvantage of Bus topology?

A. Easy to add devices

B. Low cost

C. Failure of backbone cable affects entire network

D. Suitable for large networks

Ans: C If the backbone (bus) cable fails, the entire network goes down. This is a major disadvantage of Bus topology.

Q104	Characteristic	Bus Topology

Bus topology uses which type of cable termination?

A. Open ends

B. Terminators at both ends

C. Loop connection

D. Circular wiring

Ans: B Bus topology requires terminators at both ends of the backbone cable to prevent signal reflection, which would cause data corruption.

Q105	Advantage	Bus Topology

What is an advantage of Bus topology?

A. High fault tolerance

B. Easy to troubleshoot

C. Low cost and easy to install for small networks

D. Best performance under heavy traffic

Ans: C Bus topology is inexpensive and easy to install for small networks — only one cable is needed to connect all devices.

Q106	Definition	Star Topology

In a Star topology:

A. All devices are on a single bus

B. All devices connect to a central hub or switch

C. Devices form a closed loop

D. Devices connect to each other directly

Ans: B In Star topology, all devices (nodes) connect to a central hub or switch. All communication goes through the central device.

Q107	Advantage	Star Topology

Which is an advantage of Star topology?

A. Failure of central hub has no effect

B. Easy fault isolation — one cable failure does not affect others

C. Cheaper than bus topology

D. No need for a central device

Ans: B In Star topology, if one cable or device fails, only that device is affected — other connections remain active, making fault isolation easy.

Q108	Disadvantage	Star Topology

What is the main disadvantage of Star topology?

A. Difficult to add devices

B. Central hub failure affects all devices

C. Low data transfer rate

D. High cable failure rate

Ans: B If the central hub/switch fails, the entire network goes down. This is the critical vulnerability of Star topology.

Q109	Application	Star Topology

Star topology is most commonly used in:

A. Early Ethernet bus networks

B. Modern LAN implementations

C. Wide Area Networks

D. Ring networks

Ans: B Star topology using a central switch is the most widely used topology in modern LAN implementations due to ease of management and fault isolation.

Q110	Definition	Ring Topology

In a Ring topology:

A. Devices share a single bus

B. Each device connects to two neighboring devices forming a closed loop

C. All devices connect to a central hub

D. Each device connects to every other device

Ans: B In Ring topology, each device is connected to exactly two others, forming a circular (ring) path for data to travel.

Q111	Concept	Ring Topology

In a Ring topology, data travels:

A. In both directions simultaneously

B. Only in one direction (unidirectional)

C. In random directions

D. Through the central hub only

Ans: B In a standard (single) Ring topology, data travels in one direction (unidirectional/clockwise), passing through each device until it reaches the destination.

Q112	Concept	Ring Topology

Token passing in Ring topology ensures:

A. That all devices get equal bandwidth

B. That only the device holding the token can transmit

C. That data travels in both directions

D. That failed nodes are bypassed automatically

Ans: B Token Ring uses token passing — a special frame (token) circulates; only the device holding the token can transmit, preventing collisions.

Q113	Disadvantage	Ring Topology

A disadvantage of Ring topology is:

A. Expensive cabling

B. A single node failure can disrupt the entire network (single ring)

C. Cannot support token passing

D. High installation complexity only

Ans: B In a single ring, if one node or connection fails, the entire ring can be disrupted as the closed loop is broken.

Q114	Definition	Mesh Topology

In a Mesh topology:

A. All devices connect to a central hub

B. Devices form a ring

C. Every device is connected to every other device (full mesh)

D. Devices share a bus

Ans: C Full Mesh topology provides a direct link between every pair of devices, ensuring maximum redundancy and fault tolerance.

Q115	Fact	Mesh Topology

For n devices in a full mesh topology, the number of direct links required is:

A. n

B. n-1

C. n(n-1)/2

D. n^2

Ans: C Full mesh requires n(n-1)/2 links for n nodes. For example, 4 nodes need 4×3/2 = 6 direct links.

Q116	Advantage	Mesh Topology

The main advantage of Mesh topology is:

A. Low cost

B. Easy installation

C. High fault tolerance and redundancy

D. Simple management

Ans: C Mesh topology offers maximum redundancy — if one path fails, data can be rerouted through alternate paths, making it highly fault tolerant.

Q117	Disadvantage	Mesh Topology

The main disadvantage of Mesh topology is:

A. Low reliability

B. High cost due to excessive cabling and complexity

C. Low bandwidth

D. Single point of failure

Ans: B Full Mesh requires a large number of cables and ports (n(n-1)/2 connections), making it expensive and complex to install and manage.

Q118	Definition	Tree Topology

Tree topology is also known as:

A. Star-Bus topology

B. Ring-Mesh topology

C. Hierarchical topology

D. Bus-Ring topology

Ans: C Tree topology (also called hierarchical topology) arranges nodes in a tree-like structure with a root node and branching child nodes.

Q119	Characteristic	Tree Topology

Tree topology is a combination of:

A. Bus and Ring

B. Star and Bus

C. Ring and Star

D. Mesh and Bus

Ans: B Tree topology combines Star and Bus topologies — groups of star-configured devices are connected via a common bus backbone.

Q120	Definition	Hybrid Topology

Hybrid topology is:

A. A single standard topology

B. A combination of two or more different topologies

C. Used only in ring networks

D. Always cheaper than other topologies

Ans: B Hybrid topology combines two or more different topologies (e.g., star-ring, star-bus) to meet specific network requirements.

Q121	Comparison	Network Topology

Which topology requires the least amount of cable?

A. Mesh

B. Star

C. Bus

D. Ring

Ans: C Bus topology requires the least cable — all devices connect to a single backbone cable, unlike Star (separate cable per device) or Mesh (links between all pairs).

Q122	Comparison	Network Topology

Which topology has the highest fault tolerance?

A. Bus

B. Star

C. Ring

D. Mesh

Ans: D Mesh topology has the highest fault tolerance because data can be rerouted through multiple alternative paths if one link fails.

Q123	Analytical	Network Topology

In which topology is adding a new device easiest without disrupting the network?

A. Ring

B. Bus

C. Mesh

D. Star

Ans: D In Star topology, adding a new device only requires connecting it to the central hub/switch, without disrupting existing connections.

Q124	Scenario	Network Topology

A small home network where 4 computers connect to a central Wi-Fi router best represents which topology?

A. Bus

B. Mesh

C. Ring

D. Star

Ans: D A home Wi-Fi network with devices connecting to a central router is a Star topology — all traffic passes through the central device (router).

Q125	Tricky	Network Topology

Which topology was used by original Ethernet (10Base5)?

A. Star

B. Ring

C. Bus

D. Mesh

Ans: C Original Ethernet (10Base5 — thick coax, and 10Base2 — thin coax) used Bus topology with all devices sharing a single coaxial backbone.

Q126	Fact	Network Topology

In a Star topology with 10 devices, how many cables are needed?

A. 45

B. 9

C. 10

D. 5

Ans: C In Star topology, each device needs exactly one cable connecting it to the central hub/switch. So 10 devices = 10 cables.

Q127	Analytical	Network Topology

In which topology does each device act as a repeater?

A. Star

B. Bus

C. Ring

D. Mesh

Ans: C In Ring topology, each device receives data from one neighbor and retransmits (repeats) it to the next, effectively acting as a repeater.

Q128	Application	Network Topology

Which topology is most suitable for a college campus network?

A. Bus

B. Ring

C. Hybrid or Tree

D. Simple Star

Ans: C A college campus has multiple buildings and departments — Hybrid or Tree topology provides the needed scalability and hierarchy.

Q279	Concept	Network Topology

In a fully connected (full mesh) network with 6 nodes, how many links are required?

A. 6

B. 12

C. 15

D. 30

Ans: C Full mesh links = n(n-1)/2 = 6×5/2 = 15 links. Each of the 6 nodes needs a direct link to every other 5 nodes, but each link is shared, so we divide by 2.

Q280	Comparison	Network Topology

Which topology provides the best performance under heavy network load?

A. Bus

B. Star (switched)

C. Ring

D. Mesh

Ans: B Star topology with a switch provides excellent performance under heavy load — each device has a dedicated switch port (its own collision domain) eliminating competition for shared media.

Q281	Fact	Network Topology

Token Ring (IEEE 802.5) uses which topology?

A. Physical Bus, Logical Ring

B. Physical Ring, Logical Ring

C. Physical Star, Logical Star

D. Physical Mesh, Logical Ring

Ans: A Token Ring (IEEE 802.5) uses a physical Star topology (cables connect to a central MAU — Multistation Access Unit) but operates as a logical Ring (token passes in a ring pattern).

Q282	Application	Network Topology

Which topology is most commonly used in telephone networks?

A. Bus

B. Mesh (partial)

C. Ring

D. Star

Ans: D Telephone networks use Star topology — all phones connect to a central exchange (switching office). All calls are routed through the central switch.

Q283	Concept	Network Topology

In a partial mesh topology:

A. Every node is connected to every other node

B. Only some nodes have multiple connections providing redundancy

C. All nodes share a single bus

D. Nodes are arranged in a ring

Ans: B Partial mesh connects only some nodes with multiple paths (not all possible connections), balancing cost and redundancy. Used in the internet backbone.

Q284	Analytical	Network Topology

The main reason Bus topology is rarely used in modern networks is:

A. Too expensive

B. Difficult to add devices

C. Single cable failure brings down the entire network and poor scalability

D. Too fast for most applications

Ans: C Bus topology has a critical weakness: a single break in the backbone cable disrupts the entire network. It also performs poorly as more devices are added (more collisions).

Q285	Concept	Network Topology

FDDI (Fiber Distributed Data Interface) uses which topology for fault tolerance?

A. Single Ring

B. Dual Ring

C. Star

D. Mesh

Ans: B FDDI uses a Dual Ring topology — a primary ring for normal operation and a secondary ring that activates automatically if the primary ring fails, providing fault tolerance.

Q286	Fact	Network Topology

In Bus topology, what is the purpose of a terminator?

A. To amplify signals

B. To absorb signals and prevent signal reflection at cable ends

C. To connect cables

D. To provide power to devices

Ans: B Terminators are resistors placed at each end of a bus cable to absorb the electrical signal when it reaches the end, preventing signal reflection (bounce-back) that would corrupt data.

Q287	Comparison	Network Topology

Which statement correctly compares Star and Bus topology?

A. Bus uses more cable than Star

B. Star requires a central device; Bus does not

C. Star has more single points of failure

D. Bus is more fault tolerant

Ans: B Star topology requires a central hub/switch (single point of failure for the whole network). Bus topology uses no central device but is vulnerable to backbone cable failure.

Q288	Scenario	Network Topology

A network has 10 computers. In Bus topology, how many cable segments connect them (ignoring terminators)?

A. 10

B. 9

C. 1

D. 45

Ans: C In Bus topology, all 10 computers share a SINGLE backbone cable (1 cable segment). Each computer taps into this one cable — no individual cables per device like in Star topology.

Q289	Tricky	Network Topology

Which of the following is TRUE about Ring topology?

A. Data can travel in both directions simultaneously in a single ring

B. Adding a new device requires interrupting the ring

C. There is no single point of failure

D. The ring uses a central switch

Ans: B Adding a new device to a Ring requires breaking the ring to insert the new node, briefly interrupting network operation. This is a significant disadvantage of Ring topology.

Q290	Concept	Network Topology

The physical topology of a network refers to:

A. The logical flow of data

B. The actual physical layout of cables and devices

C. The protocol used

D. The number of devices

Ans: B Physical topology is the actual physical arrangement and layout of devices and cables in a network. Logical topology describes how data flows, which may differ from the physical layout.

Q291	Application	Network Topology

Which topology is best suited for a small network requiring low cost and simple installation?

A. Mesh

B. Tree

C. Bus

D. Hybrid

Ans: C Bus topology is the simplest and lowest cost for very small networks — only one backbone cable is needed. However, it doesn't scale well or provide fault tolerance.

Q292	Comparison	Network Topology

In terms of wiring cost, rank these topologies from lowest to highest: Bus, Star, Full Mesh

A. Bus < Star < Full Mesh

B. Star < Bus < Full Mesh

C. Full Mesh < Star < Bus

D. Bus < Full Mesh < Star

Ans: A Bus uses least cable (one backbone). Star uses more (one cable per device). Full Mesh uses most (n(n-1)/2 connections). So: Bus < Star < Full Mesh for wiring cost.

Chapter 5: Network Protocols

Q129 | Definition | **Network Protocols**

A network protocol is best defined as:

A. A type of network cable

B. A set of rules governing communication between devices

C. A network device

D. A type of operating system

Ans: B A protocol is a standardized set of rules that defines how data is formatted, transmitted, received, and acknowledged between communicating devices.

Q130 | Function | **Network Protocols**

Which of the following is a function of a network protocol?

A. Managing CPU speed

B. Data formatting and error detection

C. Controlling monitor resolution

D. Managing disk storage

Ans: B Network protocols handle data formatting, error detection/correction, sequencing, flow control, and addressing — ensuring reliable communication.

Q131 | Characteristic | **Network Protocols**

Which characteristic ensures protocols are understood by devices from different manufacturers?

A. Proprietary standard

B. Open / universal standard

C. Vendor-specific encoding

D. Hardware dependency

Ans: B Protocols must be open (publicly defined) standards so that devices from different manufacturers can communicate seamlessly.

Q132	Concept	Network Protocols

Which of the following is NOT a function of a network protocol?

A. Error detection

B. Flow control

C. Addressing

D. Controlling CPU voltage

Ans: D Protocols govern communication: addressing, sequencing, error detection, flow control. They do not control CPU voltage (a hardware concern).

Q133	Concept	Network Protocols

CSMA/CD stands for:

A. Carrier Sense Multiple Access with Collision Detection

B. Communication System Multiple Access with Circuit Detection

C. Carrier Signal Media Access with Collision Design

D. Circuit Switching Multiple Address with Collision Data

Ans: A CSMA/CD (Carrier Sense Multiple Access with Collision Detection) is the media access control method used in traditional Ethernet LANs.

Q134	Concept	Network Protocols

In CSMA/CD, when a collision is detected, devices:

A. Continue transmitting

B. Stop and retransmit after a random backoff time

C. Permanently stop transmitting

D. Request a new IP address

Ans: B When a collision is detected in CSMA/CD, all transmitting devices stop, wait for a random backoff period, then attempt retransmission.

Q135 Concept **Network Protocols**

Token passing as a media access protocol is used in:

A. Ethernet

B. Wi-Fi

C. Token Ring and ARCNET

D. Internet

Ans: C Token passing is used in Token Ring (IEEE 802.5) and ARCNET networks, where a special frame (token) controls which device may transmit.

Q136 Fact **Network Protocols**

The protocol that resolves IP addresses to MAC addresses is:

A. DNS

B. ARP

C. DHCP

D. ICMP

Ans: B ARP (Address Resolution Protocol) maps IP addresses (Network Layer) to MAC addresses (Data Link Layer) within a local network.

Q137 Standardization **Network Protocols**

Which body is responsible for internet protocol standards?

A. ISO

B. IEEE

C. IETF

D. ITU

Ans: C IETF (Internet Engineering Task Force) is responsible for developing and maintaining internet protocol standards, published as RFCs.

Q138	Analytical	Network Protocols

A protocol stack is:

A. A single protocol

B. A collection of protocols that work together in layers

C. A network device

D. A type of cable

Ans: B A protocol stack is a set of protocols organized in layers (like OSI or TCP/IP) that work together to enable complete network communication.

Q139	Concept	Network Protocols

Flow control in protocols is used to:

A. Speed up the sender

B. Prevent the sender from overwhelming the receiver

C. Encrypt data

D. Route packets

Ans: B Flow control prevents a fast sender from sending data faster than a slow receiver can process, avoiding data loss due to buffer overflow.

Q140	Concept	Network Protocols

Error detection in data communication is achieved using:

A. Routing tables

B. Checksums, CRC, or parity bits

C. IP addresses

D. MAC addresses

Ans: B Error detection methods include: Parity bits (simple), CRC (Cyclic Redundancy Check, used in Ethernet), and checksums (used in TCP/IP).

Q141 Tricky **Network Protocols**

Which statement about protocols is FALSE?

A. Protocols define data formatting rules

B. Protocols ensure interoperability

C. Protocols are always hardware-based

D. Protocols can be vendor-neutral standards

Ans: C Protocols are NOT always hardware-based — many protocols are software-based standards (e.g., TCP/IP, HTTP). Protocols define rules, not physical hardware.

Q142 Application **Network Protocols**

HTTP is used for:

A. File transfer

B. Email

C. Web page transfer

D. Remote login

Ans: C HTTP (HyperText Transfer Protocol) is used for transmitting web pages (HTML documents) between web servers and browsers.

Q143 Application **Network Protocols**

FTP is used for:

A. Sending email

B. Transferring files between computers

C. Web browsing

D. Remote desktop

Ans: B FTP (File Transfer Protocol) is used to transfer files between a client and server over a TCP network.

Q293	Fact	Network Protocols

CSMA/CD is defined in which IEEE standard?

A. IEEE 802.11

B. IEEE 802.3

C. IEEE 802.5

D. IEEE 802.15

Ans: B CSMA/CD is the MAC protocol defined in IEEE 802.3 (Ethernet standard). It governs how devices on a shared Ethernet segment access the transmission medium.

Q294	Concept	Network Protocols

Which of the following correctly describes the 'Carrier Sense' part of CSMA/CD?

A. Devices sense whether a carrier signal (transmission) is occurring before transmitting

B. Devices transmit without checking the medium

C. Devices use a token to transmit

D. Devices transmit in assigned time slots

Ans: A In CSMA/CD, 'Carrier Sense' means a device listens to the medium first. If it detects a carrier signal (another device is transmitting), it waits before attempting to transmit.

Q295	Concept	Network Protocols

In CSMA/CD, 'Multiple Access' means:

A. Only one device can access the medium at a time

B. Multiple devices share and can access the same transmission medium

C. Multiple protocols are used

D. Multiple cables are used

Ans: B 'Multiple Access' in CSMA/CD means multiple devices share the same communication medium (the bus or hub) and any device can attempt to transmit when the medium is idle.

Q296	Concept	Network Protocols

The 'backoff algorithm' in CSMA/CD:

A. Increases transmission speed after a collision

B. Selects a random wait time before retransmission after a collision

C. Permanently stops a device from transmitting

D. Routes packets to a different network

Ans: B After a collision, CSMA/CD uses a random backoff (wait) algorithm — each device waits a random amount of time before retransmitting, reducing the chance of another collision.

Q297	Comparison	Network Protocols

CSMA/CA (used in Wi-Fi) differs from CSMA/CD because CSMA/CA:

A. Detects collisions after they happen

B. Tries to avoid collisions before they occur

C. Does not sense the carrier

D. Is used only in wired networks

Ans: B CSMA/CA (Collision Avoidance) tries to prevent collisions before they happen (by waiting random times, using RTS/CTS handshake). CSMA/CD detects collisions after they occur.

Q298	Fact	Network Protocols

Which of the following is a connection-oriented protocol at the Transport layer?

A. UDP

B. IP

C. TCP

D. ICMP

Ans: C TCP (Transmission Control Protocol) is a connection-oriented Transport Layer protocol — it establishes a connection via 3-way handshake before data transfer and ensures reliable delivery.

Q299	Fact	Network Protocols

Which of the following is a connectionless protocol at the Transport layer?

A. TCP

B. FTP

C. UDP

D. HTTP

Ans: C UDP (User Datagram Protocol) is a connectionless Transport Layer protocol — it sends datagrams without establishing a connection or guaranteeing delivery.

Q300	Concept	Network Protocols

The concept of 'sliding window' in protocols is used for:

A. Error detection only

B. Routing

C. Flow control — allowing multiple frames to be sent before requiring acknowledgement

D. Encryption

Ans: C The sliding window protocol allows a sender to transmit multiple frames (up to a window size) before needing an acknowledgement, improving throughput and efficiency.

Q301	Application	Network Protocols

Which protocol is used to find the MAC address of a device when its IP address is known?

A. DNS

B. RARP

C. ARP

D. ICMP

Ans: C ARP (Address Resolution Protocol) is used to resolve IP addresses to MAC addresses. A device broadcasts an ARP request, and the device with that IP responds with its MAC address.

Q302 Fact **Network Protocols**

RARP (Reverse Address Resolution Protocol) is used to:

A. Find MAC address from IP address

B. Find IP address from MAC address

C. Translate domain names

D. Route packets

Ans: B RARP resolves MAC addresses to IP addresses — used by diskless workstations to discover their IP address by broadcasting their MAC address (largely replaced by DHCP).

Q303 Concept **Network Protocols**

The purpose of a 'checksum' in network protocols is:

A. To encrypt data

B. To detect errors in transmitted data

C. To route data

D. To compress data

Ans: B A checksum is a value calculated from data that is sent along with the data. The receiver recalculates the checksum and compares it to detect errors introduced during transmission.

Q304 Concept **Network Protocols**

CRC (Cyclic Redundancy Check) in Ethernet is used for:

A. Routing

B. IP addressing

C. Error detection in frames

D. Encryption

Ans: C CRC (Cyclic Redundancy Check) is the error detection method used in Ethernet frames. The FCS (Frame Check Sequence) field contains the CRC value calculated over the frame contents.

Q305 Analytical **Network Protocols**

Which of the following correctly describes the difference between a protocol and a standard?

A. They are identical

B. A protocol is a set of communication rules; a standard is an officially approved specification

C. Standards are for hardware only

D. Protocols are for wireless only

Ans: B A protocol defines the rules for communication. A standard is a protocol or specification that has been officially approved by a standards body (e.g., IEEE, ISO, IETF) for widespread use.

Q306 Tricky **Network Protocols**

In token passing, if the device holding the token has nothing to send, it:

A. Holds the token until it has data

B. Passes the token to the next device immediately

C. Destroys the token

D. Broadcasts the token

Ans: B If a device holding the token has no data to transmit, it immediately passes the token to the next device. This ensures efficient, fair access to the network for all devices.

Q307 Fact **Network Protocols**

Which IEEE standard defines the Token Ring (token passing ring) protocol?

A. IEEE 802.3

B. IEEE 802.5

C. IEEE 802.11

D. IEEE 802.15

Ans: B IEEE 802.5 defines the Token Ring standard, which uses token passing over a ring topology (physical star, logical ring using a MAU).

Chapter 6: Ethernet

Q144 Definition **Ethernet**

Ethernet is defined as:

A. A wireless networking standard

B. A wired LAN technology standard (IEEE 802.3)

C. A WAN technology

D. A routing protocol

Ans: B Ethernet is a family of wired LAN technologies standardized by IEEE 802.3, defining physical and data link layer specifications.

Q145 Fact **Ethernet**

Ethernet is defined by which IEEE standard?

A. IEEE 802.11

B. IEEE 802.3

C. IEEE 802.15

D. IEEE 802.16

Ans: B IEEE 802.3 is the standard that defines Ethernet — covering frame format, MAC, physical layer specifications, and speeds.

Q146 History **Ethernet**

Ethernet was originally developed by:

A. IBM

B. Microsoft

C. Xerox, Intel, and DEC (DIX Ethernet)

D. Cisco

Ans: C Ethernet was originally developed by Robert Metcalfe at Xerox PARC in 1973, then standardized jointly by DEC, Intel, and Xerox (DIX standard).

Q147 Characteristic **Ethernet**

The original Ethernet standard operated at:

A. 1 Mbps

B. 10 Mbps

C. 100 Mbps

D. 1 Gbps

Ans: B The original IEEE 802.3 Ethernet standard operated at 10 Mbps (10Base5 using thick coaxial cable, later 10Base2 and 10Base-T).

Q148 Fact **Ethernet**

10Base-T Ethernet uses which type of cable?

A. Coaxial cable

B. Fiber optic cable

C. Twisted pair cable (Cat3/Cat5)

D. Wireless

Ans: C 10Base-T (10 Mbps Baseband, Twisted pair) uses twisted pair cables (Category 3 or higher) with RJ-45 connectors.

Q149 Fact **Ethernet**

Fast Ethernet operates at:

A. 10 Mbps

B. 100 Mbps

C. 1 Gbps

D. 10 Gbps

Ans: B Fast Ethernet (IEEE 802.3u) operates at 100 Mbps — 10 times faster than original Ethernet. Examples: 100Base-TX, 100Base-FX.

Q150 Fact **Ethernet**

Gigabit Ethernet operates at:

A. 100 Mbps

B. 1 Gbps

C. 10 Gbps

D. 100 Gbps

Ans: B Gigabit Ethernet (IEEE 802.3ab/802.3z) operates at 1 Gbps (1000 Mbps), widely used in modern enterprise LAN backbones and desktop connections.

Q151 Concept **Ethernet**

Ethernet uses which media access control method?

A. Token passing

B. TDMA

C. CSMA/CD

D. FDMA

Ans: C Traditional Ethernet uses CSMA/CD (Carrier Sense Multiple Access with Collision Detection) for media access control in shared-medium environments.

Q152 Concept **Ethernet**

In Ethernet, what is a collision domain?

A. A network segment where collisions can occur

B. A domain used for routing

C. A security zone

D. A type of IP address range

Ans: A A collision domain is a network segment where data packets can collide — in traditional (hub-based) Ethernet, all ports share one collision domain.

Q153 Concept **Ethernet**

A switch reduces collisions in Ethernet by:

A. Increasing cable length

B. Creating separate collision domains for each port

C. Removing the need for cables

D. Using token passing

Ans: B An Ethernet switch creates a separate collision domain for each port, eliminating collisions between ports and significantly improving performance.

Q154 Feature **Ethernet**

An Ethernet frame includes:

A. Source and destination MAC addresses, type/length, data, and FCS

B. Only IP addresses

C. Only data payload

D. Source and destination port numbers only

Ans: A An Ethernet frame contains: Preamble, Destination MAC, Source MAC, Type/Length, Data payload, and FCS (Frame Check Sequence for error detection).

Q155 Fact **Ethernet**

The maximum size of an Ethernet frame data field (payload) is:

A. 64 bytes

B. 1500 bytes

C. 9000 bytes

D. 256 bytes

Ans: B Standard Ethernet maximum transmission unit (MTU) is 1500 bytes for the data payload. Jumbo frames can go up to 9000 bytes.

Q156 Fact **Ethernet**

The minimum size of an Ethernet frame is:

A. 14 bytes

B. 32 bytes

C. 64 bytes

D. 128 bytes

Ans: C The minimum Ethernet frame size is 64 bytes (including headers and FCS). Frames smaller than 64 bytes are considered 'runts' and are discarded.

Q157 Advantage **Ethernet**

Which is an advantage of Ethernet?

A. Very expensive

B. Low performance

C. Cost-effective, widely supported, and easy to install

D. Works only with one operating system

Ans: C Ethernet is inexpensive, widely available, highly compatible, standards-based, and easy to install, making it the most popular LAN technology.

Q158 Limitation **Ethernet**

A limitation of traditional (hub-based) Ethernet with CSMA/CD is:

A. Cannot carry data

B. Performance degrades under heavy traffic due to collisions

C. Does not support IP

D. Only works on fiber

Ans: B In shared Ethernet (hub-based), as traffic increases, collisions increase, causing devices to wait and retransmit, reducing overall throughput.

Q159	Concept	Ethernet

10Base2 Ethernet (Thinnet) uses which type of cable?

A. Twisted pair

B. Thin coaxial cable

C. Fiber optic

D. Thick coaxial

Ans: B 10Base2 (Thinnet) uses thin coaxial cable (RG-58), operates at 10 Mbps with a maximum segment length of 185 metres.

Q160	Comparison	Ethernet

Full-duplex Ethernet eliminates collisions because:

A. It uses token passing

B. It uses separate transmit and receive channels simultaneously

C. It limits network to one user

D. It uses CSMA/CD more aggressively

Ans: B Full-duplex Ethernet uses separate wires/channels for sending and receiving simultaneously. With no shared medium, CSMA/CD is not needed and collisions cannot occur.

Q161	Tricky	Ethernet

What does the 'Base' in 10Base-T refer to?

A. The base of the cable

B. Baseband transmission (single channel using entire bandwidth)

C. Base station

D. Basic transmission

Ans: B In Ethernet naming (e.g., 10Base-T), '10' = speed in Mbps, 'Base' = Baseband (entire bandwidth used for one channel), 'T' = Twisted pair cable.

Q162	Fact	Ethernet

IEEE 802.3ab is the standard for:

A. 10 Mbps Ethernet

B. Fast Ethernet (100 Mbps)

C. Gigabit Ethernet over twisted pair (1000Base-T)

D. 10 Gigabit Ethernet

Ans: C IEEE 802.3ab defines 1000Base-T (Gigabit Ethernet over 4-pair Cat5e/Cat6 UTP cable), the most common form of Gigabit Ethernet today.

Q163	Concept	Ethernet

A broadcast domain in Ethernet is a segment where:

A. Only two devices can communicate

B. A broadcast frame is received by all devices

C. No frames are forwarded

D. Only routers communicate

Ans: B A broadcast domain is a network segment where a broadcast frame (addressed to FF:FF:FF:FF:FF:FF) is received by all devices. Routers separate broadcast domains.

Q164	Comparison	Ethernet

A switch divides collision domains but NOT broadcast domains. What device divides both?

A. Hub

B. Repeater

C. Switch

D. Router

Ans: D A router divides both collision domains AND broadcast domains. A switch divides collision domains but all ports typically share one broadcast domain.

Q165	Fact	Ethernet

The preamble in an Ethernet frame is used for:

A. Error detection

B. Destination addressing

C. Bit synchronization between sender and receiver

D. Data payload

Ans: C The 7-byte preamble (alternating 1s and 0s) allows the receiver's hardware to synchronize its clock with the incoming signal before reading the actual frame data.

Q308	Fact	Ethernet

Which Ethernet variant uses fiber optic cable and is designed for long-distance connections?

A. 10Base-T

B. 10Base2

C. 100Base-FX

D. 1000Base-T

Ans: C 100Base-FX is a Fast Ethernet variant using fiber optic cable, supporting 100 Mbps over longer distances than copper-based variants (up to 412 metres half-duplex, 2 km full-duplex).

Q309	Concept	Ethernet

In Ethernet, the SFD (Start Frame Delimiter) is used to:

A. Indicate the end of the frame

B. Mark the beginning of the actual frame after the preamble

C. Provide error detection

D. Indicate the frame type

Ans: B The SFD (Start Frame Delimiter) is a 1-byte field (10101011) that follows the preamble and marks the exact beginning of the Ethernet frame data.

Q310 Fact **Ethernet**

The Ethernet II frame format uses which field to indicate the higher-layer protocol type?

A. Length field

B. FCS field

C. EtherType field

D. Preamble field

Ans: C The EtherType field (2 bytes) in an Ethernet II frame indicates the upper-layer protocol carried in the payload — e.g., 0x0800 = IPv4, 0x0806 = ARP, 0x86DD = IPv6.

Q311 Concept **Ethernet**

The FCS (Frame Check Sequence) in Ethernet is calculated using:

A. MD5

B. SHA-1

C. CRC-32 (Cyclic Redundancy Check)

D. Parity bits

Ans: C Ethernet uses CRC-32 for the FCS — a 4-byte value calculated over the frame contents. The receiver recalculates and compares to detect transmission errors.

Q312 Fact **Ethernet**

In 10Base-T Ethernet, what does the '10' represent?

A. Cable length in metres

B. Data rate in Mbps (10 Mbps)

C. Number of devices

D. Year of introduction

Ans: B In Ethernet naming convention, the number prefix represents the data rate in Mbps. '10' = 10 Mbps, '100' = 100 Mbps (Fast Ethernet), '1000' = 1 Gbps (Gigabit Ethernet).

Q313 Concept **Ethernet**

What is the purpose of the 'jam signal' in CSMA/CD Ethernet?

A. To reserve the medium

B. To notify all devices that a collision has occurred, so they stop transmitting

C. To start a new transmission

D. To route data

Ans: B When a collision is detected, the devices involved send a jam signal (32-bit sequence) to ensure all devices on the network know a collision occurred and should stop transmitting.

Q314 Fact **Ethernet**

10 Gigabit Ethernet (10GbE) is defined by which IEEE standard?

A. IEEE 802.3ab

B. IEEE 802.3ae

C. IEEE 802.3u

D. IEEE 802.3z

Ans: B IEEE 802.3ae defines 10 Gigabit Ethernet (10GbE), providing 10 Gbps data rates primarily used in network backbones, data centers, and high-speed server connections.

Q315 Concept **Ethernet**

Half-duplex Ethernet uses CSMA/CD because:

A. It always transmits in both directions

B. Devices share the medium and can transmit in only one direction at a time, causing potential collisions

C. It is faster than full-duplex

D. It uses token passing

Ans: B Half-duplex Ethernet shares a medium where only one device can transmit at a time. CSMA/CD manages access to prevent collisions and resolve them when they occur.

Q316	Application	Ethernet

Where is 10 Gigabit Ethernet (10GbE) most commonly used?

A. Home networks

B. Office desktops

C. Data center backbones and server connections

D. Small branch offices

Ans: C 10GbE is primarily used in high-performance environments: data center backbone connections, server-to-switch links, storage area networks, and enterprise core networks.

Q317	Tricky	Ethernet

In Ethernet, a 'runt' frame is one that:

A. Is too large (over 1518 bytes)

B. Is too small (under 64 bytes)

C. Has a bad FCS

D. Has no destination address

Ans: B A runt frame is an Ethernet frame smaller than the minimum size of 64 bytes. Runts are typically caused by collisions or faulty hardware and are discarded by receiving devices.

Q318	Fact	Ethernet

Which of the following Ethernet standards uses short-range fiber optic cable for Gigabit Ethernet?

A. 1000Base-T

B. 1000Base-SX

C. 1000Base-LX

D. 1000Base-CX

Ans: B 1000Base-SX uses multimode fiber optic cable for short-range Gigabit Ethernet (up to 550 metres). 1000Base-LX is for longer distances using single-mode or multimode fiber.

Q319	Concept	Ethernet

The MAC address in Ethernet is:

A. A 32-bit logical address

B. A 48-bit (6-byte) hardware address burned into the NIC

C. An IP address

D. A port number

Ans: B A MAC address is a 48-bit (6-byte) unique hardware address assigned to a network interface card (NIC) by the manufacturer. It is used for local network communication.

Q320	Concept	Ethernet

In Ethernet, which part of the MAC address identifies the manufacturer?

A. Last 3 bytes (OUI)

B. First 3 bytes (OUI — Organizationally Unique Identifier)

C. All 6 bytes

D. The first byte only

Ans: B The first 3 bytes of a MAC address form the OUI (Organizationally Unique Identifier) assigned by IEEE to identify the manufacturer/vendor of the network interface.

Q321	Analytical	Ethernet

Why does a switch maintain a MAC address table?

A. To assign IP addresses

B. To map port numbers to IP addresses

C. To map MAC addresses to switch ports for efficient frame forwarding

D. To encrypt frames

Ans: C A switch builds a MAC address table (CAM table) by learning which MAC address is connected to each port. This allows it to forward frames directly to the correct port instead of broadcasting.

Q322	Fact	Ethernet

Cat5e cable supports Gigabit Ethernet (1000Base-T) up to a distance of:

A. 55 metres

B. 185 metres

C. 100 metres

D. 500 metres

Ans: C Cat5e (enhanced Cat5) supports Gigabit Ethernet (1000Base-T) up to 100 metres. Cat6 also supports 100 metres for 1 Gbps and 55 metres for 10 Gbps.

Chapter 7: ARCNET

Q166 Definition **ARCNET**

ARCNET stands for:

A. Attached Resource Computer NETwork

B. Advanced Ring Communication NETwork

C. Automated Remote Computer Network

D. Access Resource Control NETwork

Ans: A ARCNET stands for Attached Resource Computer NETwork. It was one of the earliest LAN technologies developed in the late 1970s.

Q167 History **ARCNET**

ARCNET was developed by:

A. IBM

B. Xerox

C. Datapoint Corporation

D. Microsoft

Ans: C ARCNET was developed by Datapoint Corporation in 1976. It was one of the first commercially available local area networking technologies.

Q168 Fact **ARCNET**

The original data transfer rate of ARCNET was:

A. 10 Mbps

B. 100 Mbps

C. 2.5 Mbps

D. 56 Kbps

Ans: C Original ARCNET operated at 2.5 Mbps. Later versions (ARCNET Plus) increased to 20 Mbps.

Q169	Concept	ARCNET

ARCNET uses which media access control method?

A. CSMA/CD

B. CSMA/CA

C. Token passing

D. TDMA

Ans: C ARCNET uses token passing for media access control — a token circulates among devices, and only the device holding the token can transmit.

Q170	Characteristic	ARCNET

ARCNET can operate on which topology/topologies?

A. Only Bus

B. Only Star

C. Only Ring

D. Both Bus and Star topologies

Ans: D ARCNET is flexible — it can operate on both Bus and Star topologies, which contributed to its adaptability in various network environments.

Q171	Fact	ARCNET

ARCNET uses which type of cable?

A. Fiber optic only

B. Thick coaxial cable (RG-62)

C. Twisted pair only

D. Thin coaxial only

Ans: B Traditional ARCNET uses RG-62 coaxial cable (93-ohm impedance), though later versions supported twisted pair and fiber optic cables.

Q172	Concept	ARCNET

In ARCNET, node addresses are:

A. Dynamically assigned

B. Manually configured (1 to 255)

C. Assigned by a DHCP server

D. Based on IP addresses

Ans: B ARCNET node addresses are manually configured by the network administrator, ranging from 1 to 255 (maximum of 255 nodes per network).

Q173	Fact	ARCNET

The maximum number of nodes supported by ARCNET is:

A. 16

B. 64

C. 255

D. 512

Ans: C ARCNET supports a maximum of 255 nodes per network, as node addresses range from 1 to 255.

Q174	Advantage	ARCNET

An advantage of ARCNET over early Ethernet was:

A. Higher speed

B. Deterministic performance due to token passing

C. Lower cost

D. Greater maximum distance

Ans: B ARCNET's token passing provides deterministic (predictable) performance — unlike CSMA/CD Ethernet, transmission time can be precisely calculated.

Q175 Application **ARCNET**

ARCNET was commonly used in:

A. Internet backbone connections

B. Industrial automation and process control systems

C. Satellite communications

D. Mobile phone networks

Ans: B ARCNET found strong adoption in industrial automation, process control, and embedded systems due to its deterministic, reliable communication.

Q176 Limitation **ARCNET**

A limitation of ARCNET is:

A. No support for star topology

B. Slower maximum speed compared to Ethernet

C. Cannot use coaxial cable

D. No token passing support

Ans: B ARCNET's original 2.5 Mbps speed was much lower than Ethernet's 10 Mbps, which contributed to its decline as networks demanded higher bandwidth.

Q177 Comparison **ARCNET**

Compared to Ethernet, ARCNET:

A. Has higher data rate

B. Uses CSMA/CD

C. Has deterministic access but lower speed

D. Supports more nodes

Ans: C ARCNET is deterministic (token passing guarantees access time) but has lower speed (2.5 Mbps) compared to Ethernet (10+ Mbps).

Q178	Fact	ARCNET

ARCNET Plus improved the speed to:

A. 10 Mbps

B. 20 Mbps

C. 100 Mbps

D. 5 Mbps

Ans: B ARCNET Plus (an enhanced version) increased the data rate to 20 Mbps while maintaining backward compatibility with standard ARCNET.

Q179	Tricky	ARCNET

Why is ARCNET considered suitable for industrial environments?

A. Because it uses Wi-Fi

B. Because of its deterministic token-passing and robust, simple design

C. Because it operates at 10 Gbps

D. Because it uses fiber optic only

Ans: B ARCNET's token-passing mechanism guarantees deterministic, predictable communication, and its simple robust design makes it ideal for industrial control systems.

Q180	Fact	ARCNET

ARCNET operates on which impedance coaxial cable?

A. 50-ohm (RG-58)

B. 75-ohm (RG-6)

C. 93-ohm (RG-62)

D. 100-ohm (STP)

Ans: C ARCNET uses 93-ohm RG-62 coaxial cable, different from the 50-ohm cable used by Ethernet (10Base2) and 75-ohm used by cable TV.

Q181 Analytical **ARCNET**

ARCNET was eventually replaced in office environments primarily because of:

A. Its use of token passing

B. Its lower data rates compared to Ethernet and lack of widespread software support

C. Its support for star topology

D. Its use of coaxial cable

Ans: B ARCNET's 2.5 Mbps speed was insufficient as networks grew, and Ethernet's greater speed, lower cost, and broader software support led to ARCNET's replacement.

Q182 Concept **ARCNET**

In ARCNET, the token is passed to:

A. The node with the lowest address

B. The node with the next higher address (or back to 1 after 255)

C. A random node

D. The central hub only

Ans: B In ARCNET, the token passes to the next higher node address. After the highest (255), it wraps back to the lowest active node address, creating a logical ring.

Q183 Analytical **ARCNET**

Why is ARCNET's token passing considered deterministic?

A. Because it is random

B. Because the maximum wait time for a token can be precisely calculated

C. Because it uses CSMA/CD

D. Because all nodes transmit simultaneously

Ans: B Token passing is deterministic — since the token passes predictably between nodes in order, the maximum time any node must wait to transmit can be calculated exactly.

Q323	Concept	ARCNET

ARCNET's token passing creates a logical:

A. Bus

B. Star

C. Ring

D. Mesh

Ans: C Although ARCNET can use Bus or Star physical topologies, its token passing mechanism creates a logical ring — the token passes from node to node in numerical address order.

Q324	Fact	ARCNET

The cable impedance of ARCNET (RG-62) is:

A. 50 ohms

B. 75 ohms

C. 93 ohms

D. 100 ohms

Ans: C ARCNET uses RG-62 coaxial cable with a characteristic impedance of 93 ohms, different from Ethernet's RG-58 (50 ohms) and cable TV (75 ohms).

Q325	Concept	ARCNET

In ARCNET, what happens when a node wants to transmit?

A. It transmits immediately

B. It waits to receive the token, then transmits its data

C. It broadcasts a request

D. It uses CSMA/CD

Ans: B In ARCNET, a node must wait until it receives the token. Only then can it transmit its data. After transmitting (or if it has nothing to send), it passes the token to the next node.

Q326 Fact **ARCNET**

A passive hub in ARCNET can have a maximum of how many ports?

A. 4

B. 8

C. 16

D. 32

Ans: A A passive hub in ARCNET supports up to 4 ports. An active hub supports up to 8 ports and regenerates the signal, allowing longer distances between nodes.

Q327 Comparison **ARCNET**

ARCNET Plus (20 Mbps) compared to standard ARCNET (2.5 Mbps):

A. Is not backward compatible

B. Is 8 times faster and fully backward compatible

C. Uses different token passing method

D. Requires fiber optic cable only

Ans: B ARCNET Plus runs at 20 Mbps (8x faster than standard 2.5 Mbps) and is fully backward compatible with standard ARCNET devices, allowing mixed-speed networks.

Q328 Analytical **ARCNET**

The main reason ARCNET is preferred over Ethernet in certain industrial control systems is:

A. Lower cost

B. Deterministic timing — exact calculation of maximum access delay

C. Higher speed

D. Better support for IP

Ans: B Industrial control systems often require deterministic (predictable) timing for real-time control. ARCNET's token passing guarantees a maximum wait time, unlike CSMA/CD's random delays.

Q329	Fact	ARCNET

In ARCNET, node address 0 is:

A. Used for the gateway

B. Reserved — not valid for network nodes

C. The default address

D. Used for the server

Ans: B ARCNET node addresses range from 1 to 255. Address 0 is reserved and cannot be assigned to any network node.

Q330	Concept	ARCNET

Which of the following statements about ARCNET is TRUE?

A. ARCNET uses CSMA/CD for media access

B. ARCNET supports a maximum of 1024 nodes

C. ARCNET uses token passing and supports up to 255 nodes

D. ARCNET was developed by Xerox

Ans: C ARCNET uses token passing (not CSMA/CD) and supports a maximum of 255 nodes per network (addresses 1-255). It was developed by Datapoint Corporation, not Xerox.

Chapter 8: TCP/IP

Q184 Definition **TCP/IP**

TCP/IP stands for:

A. Transmission Control Protocol / Internet Protocol

B. Transfer Communication Protocol / Internet Protocol

C. Transmission Channel Protocol / Internal Protocol

D. Transfer Control Process / Internet Package

Ans: A TCP/IP stands for Transmission Control Protocol / Internet Protocol. It is the foundational communication suite of the internet.

Q185 Concept **TCP/IP**

The TCP/IP model has how many layers?

A. 7

B. 5

C. 4

D. 3

Ans: C The TCP/IP model has 4 layers: Network Access (Link), Internet, Transport, and Application layers.

Q186 Layer **TCP/IP**

The correct order of TCP/IP layers from bottom to top is:

A. Application, Transport, Internet, Network Access

B. Network Access, Internet, Transport, Application

C. Internet, Transport, Application, Network Access

D. Transport, Network Access, Internet, Application

Ans: B TCP/IP layers (bottom to top): Network Access → Internet → Transport → Application.

Q187	Function	TCP/IP

TCP (Transmission Control Protocol) provides:

A. Connectionless, unreliable service

B. Connection-oriented, reliable data delivery

C. Only routing services

D. Only addressing services

Ans: B TCP is connection-oriented and reliable — it establishes a connection (3-way handshake), ensures data delivery, and provides error correction and flow control.

Q188	Function	TCP/IP

IP (Internet Protocol) is responsible for:

A. Reliable end-to-end delivery

B. Logical addressing and routing of packets

C. Session management

D. Encryption

Ans: B IP provides logical addressing (IP addresses) and routing of packets across interconnected networks, but without reliability guarantees.

Q189	Concept	TCP/IP

UDP (User Datagram Protocol) is:

A. Connection-oriented and reliable

B. Connectionless and unreliable but faster

C. Slower than TCP

D. Used only for email

Ans: B UDP is connectionless and does not guarantee delivery, ordering, or error correction. This makes it faster than TCP, suitable for streaming/DNS/gaming.

Q190 Comparison **TCP/IP vs OSI**

How many layers does the TCP/IP model have compared to the OSI model?

A. Both have 7 layers

B. TCP/IP has 4 layers vs OSI's 7 layers

C. TCP/IP has 5 layers vs OSI's 7 layers

D. TCP/IP has 7 layers vs OSI's 4 layers

Ans: B TCP/IP has 4 layers; OSI has 7 layers. The TCP/IP Application layer combines OSI's Application, Presentation, and Session layers.

Q191 Comparison **TCP/IP vs OSI**

The Application layer of TCP/IP corresponds to which OSI layers?

A. Only the Application layer

B. Application and Presentation layers

C. Application, Presentation, and Session layers

D. Transport and Session layers

Ans: C TCP/IP's Application layer combines the OSI Application (7), Presentation (6), and Session (5) layers into one.

Q192 Comparison **TCP/IP vs OSI**

The Internet layer of TCP/IP corresponds to which OSI layer?

A. Data Link Layer

B. Transport Layer

C. Network Layer

D. Physical Layer

Ans: C The TCP/IP Internet layer corresponds to the OSI Network Layer (Layer 3) — both handle logical addressing and routing.

Q193 Comparison **TCP/IP vs OSI**

The Network Access layer of TCP/IP corresponds to which OSI layers?

A. Physical Layer only

B. Data Link Layer only

C. Physical and Data Link Layers

D. Network and Transport Layers

Ans: C The TCP/IP Network Access (Link) layer combines OSI's Physical Layer (1) and Data Link Layer (2) functions.

Q194 Concept **TCP/IP**

The TCP 3-way handshake consists of:

A. SYN, ACK, FIN

B. SYN, SYN-ACK, ACK

C. SYN, DATA, ACK

D. OPEN, SEND, CLOSE

Ans: B TCP 3-way handshake: (1) Client sends SYN, (2) Server responds SYN-ACK, (3) Client sends ACK. This establishes a reliable TCP connection.

Q195 Concept **TCP/IP**

In TCP/IP, an IP address is a:

A. Physical hardware address (MAC)

B. Logical 32-bit (IPv4) or 128-bit (IPv6) address

C. Port number

D. Protocol identifier

Ans: B An IP address is a logical address — 32 bits in IPv4 (e.g., 192.168.1.1) or 128 bits in IPv6 — used to identify devices on a network.

Q1996	Fact	TCP/IP

IPv4 addresses are:

A. 128-bit

B. 48-bit

C. 32-bit

D. 64-bit

Ans: C IPv4 addresses are 32-bit numbers, written as four decimal octets (e.g., 192.168.0.1), allowing approximately 4.3 billion unique addresses.

Q1997	Fact	TCP/IP

IPv6 addresses are:

A. 32-bit

B. 48-bit

C. 64-bit

D. 128-bit

Ans: D IPv6 uses 128-bit addresses, written as eight groups of 4 hexadecimal digits, providing a vastly larger address space than IPv4.

Q1998	Application	TCP/IP

Which of the following operates at the TCP/IP Application layer?

A. IP

B. TCP

C. HTTP

D. Ethernet

Ans: C HTTP, FTP, SMTP, DNS, and other application protocols operate at the TCP/IP Application layer, providing services to end users.

Q199 Application **TCP/IP**

Which protocol operates at the TCP/IP Internet layer?

A. HTTP

B. TCP

C. IP

D. Ethernet

Ans: C IP (Internet Protocol) operates at the Internet layer of TCP/IP, handling logical addressing and packet routing across networks.

Q200 Advantage **TCP/IP**

Which of the following is an advantage of TCP/IP?

A. Proprietary — only works with specific hardware

B. Universal, open standard supporting interoperability

C. Very slow

D. Only works on LAN

Ans: B TCP/IP is an open, universal standard that supports interoperability between different hardware, operating systems, and vendors — it is the backbone of the internet.

Q201 Feature **TCP/IP**

TCP ensures reliable delivery by using:

A. Token passing

B. Acknowledgements (ACK) and retransmission

C. CSMA/CD

D. Frame checksums only

Ans: B TCP uses acknowledgements — the receiver sends ACK for each received segment. If ACK is not received within a timeout, the sender retransmits the data.

Q202	Concept	TCP/IP

What is the role of a port number in TCP/IP?

A. Identifies the source IP address

B. Identifies the specific application/process on a host

C. Identifies the routing path

D. Identifies the cable type

Ans: B Port numbers (0-65535) identify specific applications/processes on a host. E.g., port 80 = HTTP, port 443 = HTTPS, port 25 = SMTP.

Q203	Fact	TCP/IP

The well-known port number for HTTP is:

A. 21

B. 23

C. 80

D. 443

Ans: C Port 80 is the well-known port for HTTP (unencrypted web traffic). HTTPS uses port 443, FTP uses 21, Telnet uses 23.

Q204	Fact	TCP/IP

The well-known port number for HTTPS is:

A. 80

B. 21

C. 25

D. 443

Ans: D Port 443 is used for HTTPS (HTTP Secure) — encrypted web communication using SSL/TLS.

Q205	Fact	TCP/IP

The well-known port for FTP is:

A. 20 and 21

B. 25

C. 80

D. 443

Ans: A FTP uses two ports: port 21 (control connection — commands) and port 20 (data connection — file transfer).

Q206	Concept	TCP/IP

ICMP (Internet Control Message Protocol) is used for:

A. File transfer

B. Email delivery

C. Error reporting and diagnostic messages (e.g., ping)

D. Web browsing

Ans: C ICMP operates at the Internet layer and is used for error reporting (e.g., 'destination unreachable') and diagnostic tools like ping and traceroute.

Q207	Analytical	TCP/IP

Why is UDP preferred over TCP for video streaming?

A. UDP is more reliable

B. UDP provides error correction

C. UDP is faster with no connection overhead — minor packet loss is acceptable

D. UDP uses more bandwidth

Ans: C Video streaming prioritizes speed over perfect reliability. UDP's no-connection overhead means lower latency; occasional dropped packets cause minor glitches, not unacceptable delays.

Q208	Tricky	TCP/IP

Which layer of TCP/IP is responsible for packet routing?

A. Network Access Layer

B. Application Layer

C. Transport Layer

D. Internet Layer

Ans: D The Internet Layer of TCP/IP (corresponding to OSI Network Layer) handles packet routing using IP addresses.

Q209	Fact	TCP/IP

The loopback address in IPv4 is:

A. 192.168.0.1

B. 10.0.0.1

C. 127.0.0.1

D. 172.16.0.1

Ans: C 127.0.0.1 is the loopback (localhost) address — used to test network software on the local machine without actual network communication.

Q210	Fact	TCP/IP

The default subnet mask for a Class C network is:

A. 255.0.0.0

B. 255.255.0.0

C. 255.255.255.0

D. 255.255.255.255

Ans: C Class C default subnet mask is 255.255.255.0 (/24), providing 254 usable host addresses per network (2^8 - 2 = 254).

Q211 Analytical **TCP/IP**

How many usable hosts can a /24 subnet support?

A. 256

B. 254

C. 255

D. 512

Ans: B A /24 network has 8 host bits: 2^8 = 256 addresses. Subtracting network address and broadcast address = 254 usable host addresses.

Q212 Tricky **TCP/IP**

What is the broadcast address of the network 192.168.1.0/24?

A. 192.168.1.0

B. 192.168.1.1

C. 192.168.1.254

D. 192.168.1.255

Ans: D For 192.168.1.0/24, all host bits set to 1 gives the broadcast address: 192.168.1.255. This sends a packet to all hosts on the subnet.

Q213 Application **TCP/IP**

Which port does SMTP use for sending email?

A. 25

B. 110

C. 143

D. 80

Ans: A SMTP (Simple Mail Transfer Protocol) uses port 25 for sending email between mail servers. Port 110 is POP3, port 143 is IMAP.

Q214 Assertion-Reason **TCP/IP**

Assertion: TCP is connection-oriented. Reason: TCP establishes a connection via a 3-way handshake before data transfer begins.

A. Both Assertion and Reason are correct, and Reason correctly explains Assertion

B. Both are correct but Reason does not explain Assertion

C. Assertion is correct but Reason is wrong

D. Both are wrong

Ans: A Both are correct. TCP is connection-oriented because it uses a 3-way handshake (SYN, SYN-ACK, ACK) to establish a reliable connection before data transfer.

Q215 Tricky **TCP/IP**

Which of the following uses UDP and not TCP?

A. Email (SMTP)

B. Web browsing (HTTP)

C. DNS queries

D. File transfer (FTP)

Ans: C DNS typically uses UDP (port 53) for queries because they are small, fast requests. TCP is used for zone transfers or large responses.

Q331 Concept **TCP/IP**

Which field in the IP header is used to prevent packets from circulating indefinitely on the internet?

A. Source IP address

B. Destination IP address

C. TTL (Time to Live)

D. Protocol field

Ans: C TTL (Time to Live) is a counter in the IP header, decremented by each router. When it reaches 0, the packet is discarded, preventing endless circulation of lost packets.

Q332 Fact **TCP/IP**

Which TCP/IP protocol is used by routers to exchange routing information?

A. ARP

B. ICMP

C. RIP / OSPF (routing protocols)

D. DHCP

Ans: C Routing protocols like RIP (Routing Information Protocol) and OSPF (Open Shortest Path First) are used by routers to exchange routing table information and discover network paths.

Q333 Concept **TCP/IP**

In TCP/IP, what does 'socket' refer to?

A. A physical connector

B. A combination of IP address + port number identifying a specific process on a host

C. A type of cable

D. A routing protocol

Ans: B A socket is the combination of an IP address and a port number (e.g., 192.168.1.10:80) that uniquely identifies a specific process or service on a host in a TCP/IP network.

Q334 Fact **TCP/IP**

Which of the following is a private (non-routable) IP address range?

A. 8.8.8.8

B. 200.1.1.1

C. 192.168.0.0 – 192.168.255.255

D. 150.100.1.1

Ans: C 192.168.0.0/16 is a Class C private IP address range (RFC 1918). Private IP addresses are not routed on the public internet; they are used internally in LANs.

Q335 Concept **TCP/IP**

What is the purpose of the 'sequence number' field in a TCP segment?

A. To identify the destination port

B. To number each byte of data for ordering and acknowledgement

C. To set the TTL

D. To identify the protocol

Ans: B TCP's sequence number uniquely identifies each byte of data in the stream, enabling the receiver to reorder out-of-sequence segments and the sender to detect lost data.

Q336 Fact **TCP/IP**

The well-known port for Telnet is:

A. 22

B. 23

C. 25

D. 110

Ans: B Telnet uses port 23 for remote terminal access (unencrypted). SSH (Secure Shell), the modern encrypted alternative, uses port 22.

Q337 Concept **TCP/IP**

In TCP/IP, the 'acknowledgement number' in a TCP segment indicates:

A. The sequence number of the last byte received

B. The next byte the receiver expects to receive

C. The source port number

D. The window size

Ans: B The TCP acknowledgement number specifies the next sequence number the receiver expects — essentially confirming receipt of all data up to (acknowledgement number - 1).

Q338 Analytical **TCP/IP**

Which of the following correctly describes the TCP connection termination (4-way handshake)?

A. SYN → SYN-ACK → ACK

B. FIN → ACK → FIN → ACK

C. SYN → FIN → ACK → FIN

D. ACK → FIN → SYN → ACK

Ans: B TCP connection termination: (1) Client sends FIN, (2) Server sends ACK, (3) Server sends FIN, (4) Client sends ACK. This gracefully closes both directions of the connection.

Q339 Fact **TCP/IP**

Which of the following is the correct subnet mask for a /8 network (Class A)?

A. 255.0.0.0

B. 255.255.0.0

C. 255.255.255.0

D. 255.255.255.255

Ans: A A /8 prefix (Class A) has 8 network bits, so the subnet mask is 255.0.0.0. This allows up to 16,777,214 host addresses per network.

Q340 Concept **TCP/IP**

What does 'IP fragmentation' mean?

A. Splitting an IP address

B. Dividing a large IP packet into smaller fragments to fit the MTU of a network link

C. Routing packets

D. Encrypting packets

Ans: B IP fragmentation occurs when a packet is too large for the MTU of the next network link. The router splits (fragments) the packet into smaller pieces that are reassembled at the destination.

Q341 Fact **TCP/IP**

Which of the following port numbers is used by DNS?

A. 80

B. 443

C. 53

D. 25

Ans: C DNS (Domain Name System) uses port 53 for both UDP (standard queries) and TCP (zone transfers and large responses).

Q342 Concept **TCP/IP**

What is the main function of the Internet layer in TCP/IP?

A. Providing reliable delivery

B. Physical bit transmission

C. Logical addressing (IP) and routing of packets across networks

D. Session management

Ans: C The TCP/IP Internet layer (equivalent to OSI Network Layer) provides logical addressing using IP addresses and routes packets across interconnected networks to reach the destination.

Q343 Tricky **TCP/IP**

Which protocol provides IP addresses to network devices automatically in TCP/IP networks?

A. ARP

B. ICMP

C. DHCP

D. DNS

Ans: C DHCP (Dynamic Host Configuration Protocol) automatically assigns IP addresses, subnet masks, default gateways, and DNS server addresses to network devices joining a TCP/IP network.

Q344 Analytical **TCP/IP**

A web browser making an HTTP request to a web server uses which combination of protocols?

A. UDP + IP

B. TCP + IP

C. ICMP + IP

D. UDP + Ethernet

Ans: B HTTP runs over TCP (reliable, ordered delivery) which runs over IP (routing). The full stack: HTTP (Application) → TCP (Transport) → IP (Internet) → Ethernet/Physical (Network Access).

Q345 Fact **TCP/IP**

IPv6 was introduced primarily to address which limitation of IPv4?

A. Speed limitations

B. Security limitations

C. Exhaustion of IPv4 address space (only ~4.3 billion addresses)

D. Routing complexity

Ans: C IPv4's 32-bit address space provides only ~4.3 billion unique addresses, which are nearly exhausted. IPv6 uses 128-bit addresses, providing 3.4 × 10^38 possible addresses.

Q346 Concept **TCP/IP**

What is the difference between TCP and UDP in terms of error handling?

A. Both TCP and UDP handle errors the same way

B. TCP performs error detection and recovery; UDP only performs error detection (via checksum) without recovery

C. UDP has better error handling than TCP

D. Neither TCP nor UDP handles errors

Ans: B TCP detects errors (checksum) AND recovers from them (retransmission of lost/corrupted segments). UDP has an optional checksum for detection but provides no mechanism for error recovery.

Q347 Tricky **TCP/IP**

Which protocol is used by the 'ping' command?

A. TCP

B. UDP

C. ICMP

D. ARP

Ans: C Ping uses ICMP (Internet Control Message Protocol) — specifically ICMP Echo Request messages sent to a host and ICMP Echo Reply responses, to test network connectivity.

Q348 Fact **TCP/IP**

The TCP/IP suite was originally developed for which network?

A. Ethernet

B. ARCNET

C. ARPANET (the predecessor of the internet)

D. Token Ring

Ans: C TCP/IP was originally developed for ARPANET (Advanced Research Projects Agency Network), the US Department of Defense network that was the predecessor to the modern internet.

Q349 Concept **TCP/IP**

In TCP/IP, which layer provides the interface between network services and end-user applications?

A. Internet Layer

B. Network Access Layer

C. Transport Layer

D. Application Layer

Ans: D The Application Layer is the topmost TCP/IP layer and provides the interface between the network and end-user applications (browsers, email clients, FTP programs, etc.).

Q350	Analytical	TCP/IP

Which of the following is TRUE about IPv6 compared to IPv4?

A. IPv6 has a smaller address space

B. IPv6 addresses are written in decimal

C. IPv6 provides a much larger address space and has built-in security (IPsec)

D. IPv6 is slower than IPv4

Ans: C IPv6 provides 128-bit addresses (vs. IPv4's 32-bit), an astronomically larger address space. IPv6 also has built-in support for IPsec (security), simplified headers, and auto-configuration.

Chapter 9: Advanced / Mixed Topics

Q216	Match	OSI Model

Match OSI layer with its PDU: 1-Network Layer, 2-Transport Layer, 3-Data Link Layer, 4-Physical Layer

A. 1-Frame, 2-Packet, 3-Bit, 4-Segment

B. 1-Packet, 2-Segment, 3-Frame, 4-Bit

C. 1-Segment, 2-Packet, 3-Bit, 4-Frame

D. 1-Bit, 2-Frame, 3-Segment, 4-Packet

Ans: B OSI PDUs: Network Layer → Packet | Transport Layer → Segment | Data Link Layer → Frame | Physical Layer → Bit.

Q217	Match	Network Topology

Match topology with its key characteristic: 1-Bus, 2-Star, 3-Ring, 4-Mesh

A. 1-Central hub required, 2-Single backbone, 3-Token passing, 4-All connected

B. 1-Single backbone cable, 2-Central hub required, 3-Token passing, 4-All devices connected to each other

C. 1-All connected, 2-Token passing, 3-Central hub, 4-Single backbone

D. 1-Token passing, 2-All connected, 3-Single backbone, 4-Central hub

Ans: B Bus = Single backbone cable | Star = Central hub/switch required | Ring = Token passing | Mesh = All nodes connected to each other.

Q218	Match	TCP/IP

Match TCP/IP layer with its primary protocol: 1-Application, 2-Transport, 3-Internet, 4-Network Access

A. 1-IP, 2-HTTP, 3-TCP, 4-Ethernet

B. 1-HTTP, 2-TCP / UDP, 3-IP, 4-Ethernet

C. 1-TCP, 2-IP, 3-HTTP, 4-Ethernet

D. 1-Ethernet, 2-HTTP, 3-TCP, 4-IP

Ans: B Application → HTTP/FTP/SMTP | Transport → TCP/UDP | Internet → IP/ICMP | Network Access → Ethernet/ARP.

Q219	Statement	OSI Model

Which of the following statements about the OSI model is FALSE?

A. Physical Layer transmits raw bits

B. Transport Layer provides end-to-end delivery

C. Network Layer uses MAC addresses for routing

D. Data Link Layer handles node-to-node delivery

Ans: C FALSE — the Network Layer uses IP addresses (logical addresses) for routing, NOT MAC addresses. MAC addresses are used by the Data Link Layer.

Q220	Statement	Ethernet

Which statement about Ethernet CSMA/CD is TRUE?

A. Devices transmit whenever they want

B. Devices listen before transmitting and detect collisions after

C. CSMA/CD is used in Wi-Fi

D. CSMA/CD guarantees no collisions

Ans: B CSMA/CD: Devices listen (Carrier Sense) first; if the channel is idle, they transmit; if a collision is detected during transmission, they stop and retry after a random delay.

Q221	Scenario	Network Topology

A company has 5 branch offices, each with a star-configured LAN connected via a common backbone. This represents:

A. Pure Bus topology

B. Pure Star topology

C. Tree (hierarchical) topology

D. Mesh topology

Ans: C This is Tree topology — multiple star-configured LANs (branches) are connected via a common bus backbone, forming a hierarchical tree structure.

Q222	Concept	OSI Model

The process of adding headers as data moves down the OSI layers is called:

A. Decapsulation

B. Encapsulation

C. Routing

D. Segmentation

Ans: B Encapsulation is the process of adding layer-specific headers as data moves from the Application Layer down to the Physical Layer. Decapsulation is the reverse.

Q223	Tricky	TCP/IP

A TCP segment and a UDP datagram both carry application data. The key structural difference is:

A. TCP has no port numbers

B. UDP has sequence numbers

C. TCP has sequence and acknowledgement numbers; UDP does not

D. They are identical in structure

Ans: C TCP includes sequence numbers (for ordering), acknowledgement numbers (for reliability). UDP has only source port, destination port, length, and checksum.

Q224	Concept	OSI Model

Which OSI layer deals with synchronization checkpoints during data transfer?

A. Transport Layer

B. Data Link Layer

C. Session Layer

D. Presentation Layer

Ans: C The Session Layer manages checkpoints (synchronization points) in data streams, allowing recovery from mid-transfer failures without restarting from the beginning.

Q225	Concept	TCP/IP

NAT (Network Address Translation) is used to:

A. Assign MAC addresses

B. Allow multiple private IP devices to share a public IP address

C. Encrypt data

D. Route between VLANs

Ans: B NAT translates private IP addresses (used within a LAN) to a public IP address when traffic exits to the internet, conserving public IP addresses.

Q226	Concept	Ethernet

VLAN (Virtual LAN) in Ethernet allows:

A. Wireless-only communication

B. Logical segmentation of a LAN into separate broadcast domains without separate physical hardware

C. Increased cable length

D. Only router-based communication

Ans: B VLANs logically segment a network into separate broadcast domains using the same physical switch infrastructure, improving security and reducing broadcast traffic.

Q227	Fact	Ethernet

PoE (Power over Ethernet) allows:

A. Only data transmission

B. Both data and electrical power to be delivered via Ethernet cable

C. Only power transmission

D. Wireless connectivity

Ans: B PoE (IEEE 802.3af/at) allows electrical power to be delivered alongside data over Ethernet cables, powering devices like IP phones, cameras, and access points.

Q228	Analytical	OSI Model

The gateway operates at which OSI layers?

A. Layer 1 only

B. Layer 2 only

C. Layer 3 only

D. Layers 4-7 (Transport through Application)

Ans: D A gateway operates at Layer 4 (Transport) through Layer 7 (Application) — it translates between incompatible protocols or network architectures.

Q229	Concept	Network Protocols

The maximum transmission unit (MTU) for standard Ethernet is:

A. 576 bytes

B. 1500 bytes

C. 9000 bytes

D. 512 bytes

Ans: B Standard Ethernet MTU is 1500 bytes. This is the maximum data payload size that can be carried in an Ethernet frame.

Q230 Concept **TCP/IP**

DHCP operates at which layer of the TCP/IP model?

A. Internet Layer

B. Network Access Layer

C. Transport Layer

D. Application Layer

Ans: D DHCP (Dynamic Host Configuration Protocol) is an Application Layer protocol that uses UDP (ports 67/68) to automatically assign IP addresses to hosts.

Q231 Comparison **Network Topology**

Which topology is used in the internet's backbone?

A. Bus

B. Star

C. Ring

D. Partial Mesh

Ans: D The internet backbone uses partial mesh topology — not every node is connected to every other, but multiple redundant paths exist for reliability.

Q232 Tricky **OSI Model**

SSL/TLS protocols operate primarily at which OSI layer?

A. Application Layer

B. Transport Layer

C. Presentation Layer

D. Session Layer

Ans: C SSL/TLS primarily work at the Presentation Layer (Layer 6) for encryption, though they also span Session Layer functions. They secure data in transit.

Q233 Fact **TCP/IP**

Which class of IPv4 address has the range 192.0.0.0 to 223.255.255.255?

A. Class A

B. Class B

C. Class C

D. Class D

Ans: C Class C IPv4 addresses range from 192.0.0.0 to 223.255.255.255. Class C networks support up to 254 hosts per network.

Q234 Scenario **TCP/IP**

A user sends a large file over the internet. Which protocol ensures the file arrives completely and in order?

A. UDP

B. IP

C. TCP

D. ICMP

Ans: C TCP provides reliable, ordered, complete delivery — it segments the file, numbers segments, uses ACKs, and retransmits lost segments.

Q235 Assertion-Reason **OSI Model**

Assertion: The OSI model is a reference model. Reason: It standardizes network communication regardless of vendor-specific hardware and software.

A. Both Assertion and Reason are correct, and Reason explains Assertion

B. Both are correct, but Reason is not the correct explanation

C. Assertion is correct but Reason is wrong

D. Both are wrong

Ans: A Both are correct — the OSI model is a reference model that provides a standardized framework, enabling interoperability between different vendors' products.

Q351	Comparison	OSI vs TCP/IP

The main advantage of the TCP/IP model over the OSI model is:

A. TCP/IP has more layers

B. TCP/IP is a theoretical model only

C. TCP/IP is a practical, implemented model that underlies the real internet

D. OSI is more widely used on the internet

Ans: C TCP/IP is the practical model on which the internet is built. OSI is a theoretical reference model. TCP/IP was implemented first and adopted universally, making it the real-world standard.

Q352	Concept	OSI Model

Which protocol works at BOTH the Network Layer and Transport Layer?

A. TCP

B. IP

C. UDP

D. ICMP works mainly at Network Layer; TCP/UDP at Transport Layer

Ans: D ICMP operates at the Network Layer (for error reporting between routers). TCP and UDP operate at the Transport Layer. These are distinct layers with different protocols.

Q353	Fact	Network Protocols

IEEE 802.3 and IEEE 802.11 are standards for:

A. Wired Ethernet and Wi-Fi (Wireless LAN) respectively

B. Token Ring and ARCNET

C. Both for Ethernet only

D. Both for wireless only

Ans: A IEEE 802.3 = Wired Ethernet (CSMA/CD, various speeds). IEEE 802.11 = Wi-Fi / Wireless LAN (CSMA/CA). Both are LAN standards under the IEEE 802 family.

Q354	Tricky	Basic Communication

The 'signal-to-noise ratio' (SNR) affects data communication because:

A. A higher SNR means a poorer quality signal

B. A higher SNR allows higher data rates with fewer errors

C. A lower SNR increases bandwidth

D. SNR has no effect on data rate

Ans: B A higher Signal-to-Noise Ratio (SNR) means the signal is much stronger than the noise. Higher SNR allows more signal levels to be distinguished, enabling higher data rates with fewer errors.

Q355	Scenario	TCP/IP

A company assigns IP addresses 192.168.10.1 through 192.168.10.254 to its devices. The network address is:

A. 192.168.10.255

B. 192.168.10.0

C. 192.168.0.1

D. 192.168.10.1

Ans: B 192.168.10.0 is the network address (all host bits = 0). 192.168.10.255 is the broadcast address (all host bits = 1). Usable hosts: .1 through .254.

Q356	Analytical	Ethernet

Which of the following statements about a network hub is CORRECT?

A. A hub uses MAC addresses to forward frames

B. A hub creates separate collision domains for each port

C. A hub broadcasts all incoming data to all ports

D. A hub operates at the Network Layer

Ans: C A hub (Layer 1 device) simply repeats/broadcasts any incoming signal out of all its other ports. It has no intelligence — it cannot use MAC addresses or create separate collision domains.

Q357	Concept	Network Topology

In a network, the term 'backbone' refers to:

A. The main high-speed network infrastructure connecting multiple networks or segments

B. The cables used for connecting individual devices

C. The firewall

D. The default gateway

Ans: A The backbone is the central, high-capacity part of a network that connects major segments, buildings, or network areas. It carries the bulk of network traffic.

Q358	Fact	OSI Model

Which OSI layer is responsible for managing packet sequencing?

A. Network Layer

B. Data Link Layer

C. Physical Layer

D. Transport Layer

Ans: D The Transport Layer is responsible for sequencing — numbering segments and reassembling them in the correct order at the destination, even if they arrive out of sequence.

Q359	Concept	Basic Communication

Which type of signal transmission sends data as light pulses?

A. Electrical signal transmission

B. Optical (fiber optic) transmission

C. Radio wave transmission

D. Microwave transmission

Ans: B Optical/fiber optic transmission encodes data as pulses of light (laser or LED) transmitted through glass or plastic fiber, enabling very high speeds and immunity to EMI.

Q360	Tricky	Network Protocols

Which of the following is TRUE about the Ethernet MAC address?

A. It changes when the device connects to a different network

B. It is assigned by the DHCP server

C. It is burned into the NIC by the manufacturer and is globally unique

D. It is the same as an IP address

Ans: C MAC addresses are burned into the NIC during manufacture and are globally unique (assigned by the IEEE to manufacturers via OUI blocks). They are fixed hardware addresses, unlike IP addresses.

Q361	Scenario	LAN

A university has 5 buildings, each with its own LAN. These LANs are connected by a central campus backbone. The appropriate topology for the overall campus network is:

A. Pure Bus

B. Pure Ring

C. Tree (hierarchical) or Star

D. Full Mesh

Ans: C Tree/Hierarchical topology suits a campus network — each building's LAN forms a star, and all buildings connect to the central backbone, forming the tree structure.

Q362	Comparison	TCP/IP

Which TCP/IP layer is most analogous to the OSI Transport Layer?

A. Application Layer

B. Internet Layer

C. Transport Layer

D. Network Access Layer

Ans: C The TCP/IP Transport Layer directly corresponds to the OSI Transport Layer (Layer 4). Both provide end-to-end communication services to the Application Layer above them.

Q363	Fact	Ethernet

Which of the following is TRUE about 100Base-TX Fast Ethernet?

A. It uses fiber optic cable

B. It operates at 10 Mbps

C. It uses two pairs of Cat5 or better UTP cable and operates at 100 Mbps

D. It uses coaxial cable

Ans: C 100Base-TX uses two pairs of Cat5 (or better) UTP cable with RJ-45 connectors, operating at 100 Mbps. It is the most common Fast Ethernet implementation.

Q364	Analytical	OSI Model

A firewall that filters based on IP addresses and port numbers operates at which OSI layers?

A. Layers 1-2 only

B. Layers 3-4 (Network and Transport)

C. Layers 5-7 only

D. Layer 1 only

Ans: B A packet-filtering firewall examines IP addresses (Network Layer 3) and port numbers (Transport Layer 4) to decide whether to allow or block traffic.

Q365	Concept	Basic Communication

Which of the following correctly defines 'propagation delay'?

A. The time to process a packet at a router

B. The time for a signal to physically travel from sender to receiver through the medium

C. The time to transmit all bits of a frame

D. The delay caused by queuing

Ans: B Propagation delay is the time it takes for a signal (light, electrical, radio) to physically travel from the transmitter to the receiver through the transmission medium.

Q366	Tricky	Network Topology

In a dual-ring topology (like FDDI), if the primary ring fails, the network:

A. Completely shuts down

B. Automatically wraps the traffic onto the secondary ring to continue operation

C. Requires manual reconfiguration

D. Drops all active connections permanently

Ans: B FDDI's dual-ring topology provides automatic fault recovery — if the primary ring breaks, the network wraps (loops back) to use the secondary ring, maintaining connectivity.

Q367	Fact	TCP/IP

Which class of IP address supports the largest number of networks but the fewest hosts per network?

A. Class A

B. Class B

C. Class C

D. Class D

Ans: C Class C supports 2,097,152 networks (2^21) but only 254 hosts per network (2^8 - 2). Class A supports fewer networks (128) but millions of hosts per network.

Q368	Concept	Network Protocols

Which of the following is an example of a Layer 7 (Application Layer) protocol?

A. Ethernet

B. IP

C. TCP

D. SMTP

Ans: D SMTP (Simple Mail Transfer Protocol) is an Application Layer (Layer 7) protocol. Ethernet = Layer 2, IP = Layer 3, TCP = Layer 4.

Q369	Analytical	LAN

In terms of OSI layers, a LAN primarily operates at which layers?

A. Layers 1 and 2

B. Layers 3 and 4

C. Layers 5, 6, and 7

D. All 7 layers equally

Ans: A A LAN (Local Area Network) primarily operates at the Physical Layer (Layer 1) and Data Link Layer (Layer 2). Network-level (Layer 3) functions become relevant when connecting LANs together.

Q370	Comparison	ARCNET vs Ethernet

Which of the following correctly compares ARCNET and Ethernet?

A. Both use CSMA/CD

B. ARCNET uses token passing (deterministic); Ethernet uses CSMA/CD (non-deterministic)

C. Ethernet is slower than ARCNET

D. ARCNET supports more nodes than Ethernet

Ans: B ARCNET uses deterministic token passing (predictable access times). Traditional Ethernet uses CSMA/CD (non-deterministic — access time depends on collisions and random backoff).

Q371	Concept	Basic Communication

What does 'baud rate' measure?

A. Number of bits per second

B. Number of signal changes (symbols) per second

C. Number of bytes per second

D. Number of packets per second

Ans: B Baud rate measures the number of signal changes (symbols) per second. Each symbol can represent one or more bits, so baud rate ≤ bit rate. For binary signaling, baud rate = bit rate.

Q372	Application	OSI Model

Which service does the Presentation Layer provide that is particularly important for e-commerce?

A. Routing

B. Session management

C. Encryption / SSL-TLS (converting data to a secure format)

D. Physical transmission

Ans: C The Presentation Layer's encryption function (via SSL/TLS) is critical for e-commerce — it encrypts sensitive data (credit card numbers, passwords) to protect it in transit.

Q373	Fact	TCP/IP

The 'window size' field in a TCP segment controls:

A. The number of network hops

B. The amount of data the receiver can accept before requiring an ACK (flow control)

C. The packet TTL

D. The checksum

Ans: B The TCP window size (in the segment header) tells the sender how much data (in bytes) the receiver can accept in its buffer without sending an ACK, enabling efficient flow control.

Q374	Tricky	Ethernet

An Ethernet frame with an EtherType value of 0x0800 carries which protocol?

A. ARP

B. IPv6

C. IPv4

D. VLAN tag

Ans: C EtherType 0x0800 = IPv4. 0x0806 = ARP. 0x86DD = IPv6. 0x8100 = IEEE 802.1Q VLAN-tagged frame.

Q375	Concept	Network Topology

Why is full mesh topology impractical for large networks?

A. It provides insufficient redundancy

B. The number of connections grows as n(n-1)/2 — too many cables and ports for large n

C. It is too slow

D. It cannot use token passing

Ans: B For large networks, full mesh becomes impractical: 100 nodes would need 100×99/2 = 4,950 connections. The cost, complexity, and management overhead become prohibitive.

Q376	Fact	Basic Communication

The maximum data rate of a noisy channel is given by:

A. Nyquist's theorem

B. Shannon's capacity theorem (Shannon-Hartley theorem)

C. Ohm's law

D. Moore's law

Ans: B Shannon-Hartley theorem gives the maximum data rate for a noisy channel: C = B × log■(1 + S/N), where B is bandwidth and S/N is the signal-to-noise ratio.

Q377	Concept	Network Protocols

Which of the following correctly describes a 'connectionless' protocol?

A. Establishes a dedicated path before sending data

B. Sends data without establishing a connection first — each packet is routed independently

C. Requires acknowledgement for every packet

D. Cannot detect errors

Ans: B A connectionless protocol (like IP or UDP) sends each packet independently without establishing a connection beforehand. Packets may take different routes and arrive out of order.

Q378 Analytical **TCP/IP**

A packet from PC A in London to PC B in Tokyo travels through 15 routers. At each router, the packet is processed at which layers?

A. All 7 OSI layers

B. Only Physical Layer

C. Physical, Data Link, and Network Layers (Layers 1-3)

D. Transport and Application Layers only

Ans: C At each intermediate router, the packet is received at Layer 1 (Physical), decoded at Layer 2 (Data Link — to check the frame), and routed at Layer 3 (Network — to find the next hop).

Q379 Comparison **LAN**

Which of the following correctly differentiates a switch from a router in a LAN/network context?

A. Both work at Layer 3

B. A switch connects devices within a LAN using MAC addresses (Layer 2); a router connects different networks using IP addresses (Layer 3)

C. A router is faster than a switch

D. A switch can route between subnets

Ans: B A switch operates at Layer 2 (Data Link) using MAC addresses to forward frames within a LAN. A router operates at Layer 3 (Network) using IP addresses to forward packets between different networks.

Q380 Fact **TCP/IP**

Which of the following is the correct IPv4 address range for Class B networks?

A. 1.0.0.0 – 126.255.255.255

B. 128.0.0.0 – 191.255.255.255

C. 192.0.0.0 – 223.255.255.255

D. 224.0.0.0 – 239.255.255.255

Ans: B Class B IPv4 addresses range from 128.0.0.0 to 191.255.255.255. Class B networks support 65,534 hosts per network (2^16 - 2) and 16,384 networks.

Q381	Application	Ethernet

VLANs (Virtual LANs) in a switched Ethernet network allow:

A. Physical separation of all devices

B. Logical grouping of devices into separate broadcast domains regardless of physical location

C. Wireless connectivity only

D. Direct routing between devices

Ans: B VLANs allow network administrators to logically group devices into separate broadcast domains (as if on separate physical LANs) without requiring physical separation, using VLAN-aware switches.

Q382	Concept	OSI Model

Which OSI layer is responsible for identifying communication partners?

A. Presentation Layer

B. Session Layer

C. Application Layer

D. Transport Layer

Ans: C The Application Layer (Layer 7) is responsible for identifying communication partners, determining resource availability, and synchronizing communication for end-user applications.

Q383	Tricky	TCP/IP

If a subnet has the address 172.16.5.0/24, how many usable host addresses does it have?

A. 256

B. 255

C. 254

D. 512

Ans: C A /24 subnet has 8 host bits: 2^8 = 256 addresses. Minus network address (172.16.5.0) and broadcast address (172.16.5.255) = 254 usable host addresses.

Q384	Analytical	Network Topology

The hierarchical (tree) topology is preferred in large organizations because:

A. It provides maximum redundancy

B. It is the cheapest option

C. It supports a hierarchical structure matching organizational hierarchy and is easy to expand

D. It is the fastest topology

Ans: C Tree topology matches organizational structure (corporate → departmental → workgroup), is easy to expand by adding branches, and allows centralized management at each level.

Q385	Concept	Basic Communication

Which of the following correctly describes 'data encapsulation' in network communication?

A. Removing headers from data

B. Adding protocol-specific headers and trailers at each layer as data moves down the protocol stack

C. Compressing data

D. Encrypting only the payload

Ans: B Data encapsulation is the process of wrapping data with protocol-specific control information (headers and trailers) at each layer as it passes down the protocol stack before transmission.